

Space-Time Algebra (STA)

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Abstract—Space-time algebra (STA) is developed in a paravector notation built from scalars and vectors. The algebraic development begins with the sigma basis, the conjugate and reverse operators, the decomposition of complex paravectors, dot and cross products, general paravector products, norms and scalar paravector metrics, projector algebra, exponential and logarithmic formulas, generalized transform identities, and the matrix and auxiliary realizations. The physical development then interprets the scalar paravector metric as the Minkowski metric, collects the spatial and paravector differential operators, and develops Maxwell electrodynamics together with Klein–Gordon, Dirac, and Pauli-type wave equations. Throughout, the paravector formulas are paired with explicit scalar and vector decompositions so that the component content of each result remains visible.

Index Terms—space-time algebra, geometric algebra, Lorentz transformations, Maxwell electrodynamics, Dirac equation, Pauli equation.

I. INTRODUCTION

Space-time algebra extends the familiar scalar and vector calculus of three-dimensional space into one associative algebra that can carry rotations, boosts, differential operators, and relativistic wave equations in a common language. The basic objects used here are paravectors in $\mathbb{R} \oplus \mathbb{R}^3$ and $\mathbb{C} \oplus \mathbb{C}^3$, so the same algebraic product that combines scalar and vector pieces also supplies the norm, the scalar paravector metric, the space-time interval, the electromagnetic field formulas, and the relativistic wave operators developed later in the manuscript.

The document is organized from simple algebraic structure to physical application. It begins with the sigma basis, the conjugate and reverse operators, the decomposition of a complex paravector into real scalar, imaginary scalar, real vector, and imaginary vector parts, the dot and cross products, the general paravector product laws, and the resulting norm and scalar-paravector-metric identities. It then develops projector algebra, exponential and logarithmic formulas, generalized transform identities, and the matrix and auxiliary 2×2 realizations needed later for the Dirac-type constructions. Only after that algebraic setup is fixed does the presentation turn to space-time kinematics, differential-operator notation, electromagnetism, and relativistic wave equations.

This perspective is consistent with the modern STA and paravector literature of Hestenes, Baylis, and Doran–Lasenby [12–14], while its algebraic lineage runs back through Hamilton’s quaternion construction and Clifford’s geometric extension [1, 3]. A one-page Equation Reference Sheet follows this introduction and serves as a reference sheet for the numbered formulas developed in the body.

Bold symbols denote three-vectors, and the remaining symbols are defined where they first appear. Whenever a new symbol is introduced, the text states whether it is a scalar, vector, or paravector and whether it is real or complex. Complexification is carried by the pseudoscalar i . Natural units $\hbar = c = 1$ and metric signature $(+, -, -, -)$ are used unless stated otherwise.

A. Reading Guide

1. The Equation Reference Sheet on the next page collects the main numbered equations in the order they are developed.
2. Section II develops the algebraic foundations: the sigma basis, the conjugate and reverse operators, complex paravectors, dot and cross products, paravector products, norms and scalar paravector metrics, projector algebra, exponential and logarithmic formulas, generalized transform identities, and the matrix and auxiliary 2×2 realizations.
3. Section III interprets the earlier scalar paravector metric as the Minkowski metric of interval geometry and develops proper velocity, proper momentum, acceleration, and free relativistic motion.
4. Section V collects the spatial and paravector differential operators in one place, fixing the notation for ∂_r , ∇ , ∂ , ∂^* , and $\square$ before they are used in the physical sections.
5. Section VI introduces the field and potential paravectors, Maxwell theory, gauge structure, wave equations for the field, and the Lorentz force.
6. Section VII develops the relativistic wave-equation material, from frequency splitting and Klein–Gordon factorization to the Dirac two-component field of complex paravectors, the Pauli limit, and the energy/projector split.
7. Section VIII summarizes the main structural consequences of the paravector split and records the principal directions that remain to be developed.
8. After the references, Section IX records scope notes and cautions, and Section X records verification and formula status.

Equation Reference Sheet

Natural units: $\hbar = c = 1$. Metric signature: $(+, -, -, -)$.

		Ref.
Pseudoscalar	$i = \sigma_1 \sigma_2 \sigma_3$	(1)
Sigma product	$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k$	(2)
Conjugate operator	$(\cdot)^* = \text{conjugate that flips signs of } \sigma_n$	(3)
Reverse operator	$(\cdot)^R = \text{reverse order of multiplied } \sigma_n$	(4)
Complex paravector	$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = S + \mathbf{V} \cdot \boldsymbol{\sigma} = X + iY$	(11), (17)
Real function	$\text{real}(Z) = X$	(18)
Img function	$\text{img}(Z) = Y$	(19)
Scalar function	$\text{scalar}(Z) = S$	(20)
Vector function	$\text{vector}(Z) = \mathbf{V}$	(21)
Conjugate paravector	$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$	(22)
Real conjugate	$X^* = a - \mathbf{A} \cdot \boldsymbol{\sigma}$	(25)
Reverse paravector	$Z^R = X - iY = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}$	(23)
Real reverse	$X^R = X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$	(26)
Reverse conjugate	$Z^{*R} = (a + ib) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$	(24)
Real reverse conjugate	$X^{*R} = a - \mathbf{A} \cdot \boldsymbol{\sigma} = X^*$	(27)
Vector product	$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$	(31)
Paravector product	$ZZ' = SS' + \mathbf{V} \cdot \mathbf{V}' + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}')) \cdot \boldsymbol{\sigma}$	(32)
Paravector commutator	$[Z, Z'] = 2i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}$	(33)
Scalar norm	$ S ^2 = SS^* = a^2 + b^2$	(34)
Vector norm	$\ \mathbf{V}\ ^2 = \mathbf{V} \cdot \mathbf{V}^* = \mathbf{A}^2 + \mathbf{B}^2$	(35)
Paravector norm	$\ Z\ ^2 = \text{scalar}(ZZ^R) = S ^2 + \ \mathbf{V}\ ^2$	(36)
Real scalar paravector metric	$\langle\langle X \rangle\rangle^2 = XX^* = a^2 - \mathbf{A}^2$	(38)
Scalar paravector metric	$\langle\langle Z \rangle\rangle^2 = ZZ^{*R} = S^2 - \mathbf{V}^2$	(37)
Paravector inverse	$Z^{-1} = Z^{*R} / \langle\langle Z \rangle\rangle^2 = (S - \mathbf{V} \cdot \boldsymbol{\sigma}) / (S^2 - \mathbf{V}^2)$	(42)
Projector	$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$	(45)
Functional calculus	$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+) + f(-)\Pi(-)$	(46)
Vector exponential	$\exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}$	(48)
Paravector logarithm	$\log(Y) = \frac{1}{2} \log(\lambda(+)\lambda(-)) + \frac{1}{2} \log(\lambda(+)/\lambda(-)) \mathbf{u} \cdot \boldsymbol{\sigma}$	(50)
General transform	$X' = T^{-1}XT$	(55)
Rotor	$R(\theta, \mathbf{u}) = \exp(\frac{\theta}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}), X' = R^R X R$	(52)
Boost	$L(\theta, \mathbf{u}) = \exp(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}), X' = L^* X L$	(49)
Space-time interval	$ds^2 = \langle\langle dX \rangle\rangle^2 = \langle\langle dt + \boldsymbol{\sigma} \cdot d\mathbf{r} \rangle\rangle^2 = dt^2 - d\mathbf{r}^2$	(60)
Proper velocity	$U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$	(61)
Proper momentum	$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$	(62)
Frequency operator	$\Omega = \langle\langle m + \boldsymbol{\sigma} \cdot \nabla \rangle\rangle = \sqrt{m^2 - \nabla^2}$	(87)
Frequency split	$(\partial_t \pm i\Omega)\Psi(\pm) = 0$	(87)
Spatial derivative	$\partial_{\mathbf{r}} = \{\partial/\partial r_1, \partial/\partial r_2, \partial/\partial r_3\}$	(66)
Gradient notation	$\nabla = \partial_{\mathbf{r}} = \{\partial_1, \partial_2, \partial_3\}$	(67)
Spatial Laplacian	$\nabla^2 = \partial_1^2 + \partial_2^2 + \partial_3^2$	(68)
Paravector derivative	$\partial = \partial_t + \boldsymbol{\sigma} \cdot \nabla = \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3$	(69)
Conjugate derivative	$\partial^* = \partial_t - \boldsymbol{\sigma} \cdot \nabla = \partial_t - \sigma_1 \partial_1 - \sigma_2 \partial_2 - \sigma_3 \partial_3$	(70)
d'Alembertian	$\square = \langle\langle \partial \rangle\rangle^2 = \partial \partial^* = \partial^* \partial = \partial_t^2 - \nabla^2$	(71), (72)
Field paravector	$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$	(73)
Maxwell equation	$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$	(77)
Field wave equation	$\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$	(80)
Charge conservation	$\partial_t \rho + \nabla \cdot \mathbf{J} = 0$	(81)
Lorentz force	$m dU/ds = q \text{real}(FU)$	(83)
Potential paravector	$\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$	(73)
Potential gradient	$\partial \Phi = S + F^*$	(75)
Gauge law	$\Phi' = \Phi + \partial^* \lambda$	(82)
Klein-Gordon	$(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0$	(89)
Covariant energy	$E = i\partial_t - qV$	(88)
Covariant momentum	$\mathbf{P} = -i\nabla - q\mathbf{A}$	(88)
Dirac factorization	$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$	(92)
Dirac system	$(W(i\partial - q\Phi^*) - mI)\Psi = 0$	(94)
W map	$W(Z) = \{0, Z; Z^{*R}, 0\}$	(58)
Dirac square	$\{(-E^2 + \mathbf{P}^2 + m^2 - iqF^*)\Psi_1; (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2\} = 0$	(95)
Pauli spin split	$i\partial_t \Psi(\pm) = \left((\mathbf{P}^2 \mp q\ \mathbf{B}\)/(2m) + qV \right) \Psi(\pm)$	(100)
Pauli limit	$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi$	(97)

II. ALGEBRAIC FOUNDATIONS

This section fixes the basis, the conjugate and reverse operators, complex paravectors, dot and cross products, norms and metrics, paravector products, projector formulas, exponential and logarithmic formulas, generalized transforms, and the standard matrix realizations.

Basis and pseudoscalar

For $n \in \{1, 2, 3\}$, the algebra is generated by the three basis elements σ_n . Their products generate scalars, vectors, bivectors, and pseudoscalars. The basis relations are

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ i &:= \sigma_1 \sigma_2 \sigma_3, & i^2 &= -1. \end{aligned} \quad (1)$$

The element i is a pseudoscalar because $i\sigma_n = \sigma_n i$, and squaring i returns the scalar -1 :

$$\begin{aligned} (\sigma_1 \sigma_2 \sigma_3)^2 &= \sigma_1 \sigma_2 (\sigma_3 \sigma_1) \sigma_2 \sigma_3, \\ &= -\sigma_1 (\sigma_2 \sigma_1) (\sigma_3 \sigma_2) \sigma_3, \\ &= -\sigma_1^2 \sigma_2^2 \sigma_3^2, \\ &= -1. \end{aligned}$$

With the Kronecker delta and Levi-Civita symbol,

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k. \quad (2)$$

where

$$\begin{aligned} \delta_{nm} &:= \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} \\ \epsilon_{nmk} &:= \begin{cases} +1, & (n, m, k) \text{ even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ odd permutation of } (1, 2, 3), \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

The Conjugate and Reverse Operators

The two basic operators are

$$(\cdot)^* = \text{conjugate that flips signs of } \sigma_n, \quad (3)$$

$$(\cdot)^R = \text{reverse order of multiplied } \sigma_n, \quad (4)$$

In operator form,

$$\begin{aligned} (XY)^* &= X^* Y^*, & (XY)^R &= Y^R X^R, \\ X^{*R} &= (X^R)^* = (X^*)^R. \end{aligned} \quad (5)$$

Applied to the generators and the pseudoscalar, these rules reduce to

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^R = \sigma_n, \quad \sigma_n^{*R} = -\sigma_n, \quad (6)$$

$$i^* = -i, \quad i^R = -i, \quad i^{*R} = i. \quad (7)$$

For a two-generator bivector pair with $n \neq m$,

$$\begin{aligned} (\sigma_n \sigma_m)^* &= \sigma_n^* \sigma_m^* = (-\sigma_n)(-\sigma_m) = \sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^R &= \sigma_m^R \sigma_n^R = \sigma_m \sigma_n = -\sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^{*R} &= (\sigma_m \sigma_n)^* = \sigma_m^* \sigma_n^* \\ &= (-\sigma_m)(-\sigma_n) = \sigma_m \sigma_n = -\sigma_n \sigma_m. \end{aligned} \quad (8)$$

Equation (8) records the same sign pattern on the sigma-number-two elements. These rules agree with standard STA conventions [12, 14].

The sigma number of a reduced basis product is the number of sigma generators that remain after simplification. Thus 1, σ_n , $\sigma_n \sigma_m$ with $n \neq m$, and i have sigma numbers 0, 1, 2, and 3. Any unreduced product with sigma number larger than 3 simplifies to one of these four cases. Tables 1 and 2 summarize the basis elements and operator signs by sigma number.

Table 1. Sigma basis elements by sigma number.

sigma number	representative basis elements
0 (scalar)	1
1 (vector)	$\sigma_1, \sigma_2, \sigma_3$
2 (bivector)	$\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2$
3 (pseudoscalar)	$\sigma_1 \sigma_2 \sigma_3$

Table 2. Operator signs by sigma number.

sigma number	$(\cdot)^*$	$(\cdot)^R$	$(\cdot)^{*R}$
0 (scalar)	+	+	+
1 (vector)	−	+	−
2 (bivector)	+	−	−
3 (pseudoscalar)	−	−	+

Generic Complex Paravector

Collect the three generators into the sigma-vector

$$\boldsymbol{\sigma} := \{\sigma_1, \sigma_2, \sigma_3\}, \quad (9)$$

which is a vector of algebraic objects rather than a vector of scalar components. Multiplication by the pseudoscalar gives the associated bivector triple

$$i\boldsymbol{\sigma} := \{i\sigma_1, i\sigma_2, i\sigma_3\} = \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \quad (10)$$

because $i = \sigma_1 \sigma_2 \sigma_3$ and each product $i\sigma_n$ reduces to the complementary ordered two-generator product. Equation (10) therefore records the canonical sigma-number-two basis built from the generators. With this basis fixed, for real scalars $a, b \in \mathbb{R}$ and real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, with $\mathbf{A} \cdot \boldsymbol{\sigma} = A_1 \sigma_1 + A_2 \sigma_2 + A_3 \sigma_3$, define the real paravectors $X, Y \in \mathbb{R} \oplus \mathbb{R}^3$ by

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma}. \quad (11)$$

$$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (12)$$

We can define a complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$ either as a real paravector plus an imaginary paravector with

$$Z = X + iY. \quad (13)$$

or as a scalar plus a vector component with

$$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \quad (14)$$

$$= S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (15)$$

where the complex scalar and vector components are given by

$$S = a + ib \in \mathbb{C}. \quad (16)$$

$$\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3. \quad (17)$$

Define the real, imaginary, scalar, and vector functions by

$$\text{real}(Z) := X, \quad (18)$$

$$\text{img}(Z) := Y. \quad (19)$$

$$\text{scalar}(Z) := S, \quad (20)$$

$$\text{vector}(Z) := \mathbf{V}. \quad (21)$$

For the scalar part and vector part of the paravector Z , we also write $Z_S := \text{scalar}(Z)$ and $Z_V := \text{vector}(Z)$.

The conjugate and reverse operators act on the generic complex paravector by

$$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (22)$$

$$Z^R = X - iY = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (23)$$

$$Z^{*R} = (a + ib) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (24)$$

For the real paravector X from Eq. (11), these reduce to

$$X^* = a - \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (25)$$

$$X^R = X = a + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (26)$$

$$X^{*R} = a - \mathbf{A} \cdot \boldsymbol{\sigma} = X^*. \quad (27)$$

Equations (22)–(24) are the generic operator formulas used throughout the later algebraic constructions, while Eqs. (25)–(27) record the purely real specialization used whenever the imaginary scalar and imaginary vector parts vanish. Equations (11)–(21) fix the real and complex paravector pieces and extractors used throughout the later algebra. Table 3 records the type assignments, Table 4 records the recovery formulas for the extractor definitions, and Table 5 records the same split in components.

Table 3. Type assignments for the generic complex paravector notation.

Type name	Underlying set	Example
real scalar	$\mathbb{R}$	a, b
real vector	$\mathbb{R}^3$	$\mathbf{A}, \mathbf{B}$
complex scalar	$\mathbb{C}$	S
complex vector	$\mathbb{C}^3$	$\mathbf{V}$
real paravector	$\mathbb{R} \oplus \mathbb{R}^3$	X, Y
complex paravector	$\mathbb{C} \oplus \mathbb{C}^3$	Z

Table 4. Components of Z .

scalar:	$S = (Z + Z^{R*})/2$
vector:	$\mathbf{V} \cdot \boldsymbol{\sigma} = (Z - Z^{R*})/2$
real:	$X = (Z + Z^R)/2$
img:	$Y = (Z - Z^R)/(2i)$

Table 5. Real and imaginary scalar/vector parts of Z .

real scalar:	$a = (S + S^R)/2$ $= (X + X^*)/2$
img scalar:	$b = (S - S^R)/(2i)$ $= (Y + Y^*)/2$
real vector:	$\mathbf{A} \cdot \boldsymbol{\sigma} = (\mathbf{V} + \mathbf{V}^R) \cdot \boldsymbol{\sigma}/2$ $= (X - X^*)/2$
img vector:	$\mathbf{B} \cdot \boldsymbol{\sigma} = (\mathbf{V} - \mathbf{V}^R) \cdot \boldsymbol{\sigma}/(2i)$ $= (Y - Y^*)/2$

Dot products and cross products

For real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, the dot product is defined by

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &:= \sum_{n=1}^3 \sum_{m=1}^3 \delta_{nm} A_n B_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3. \end{aligned} \quad (28)$$

Define the square of a vector by

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (29)$$

For the cross product, the Levi-Civita form gives

$$(\mathbf{A} \times \mathbf{B})_k := \sum_{n=1}^3 \sum_{m=1}^3 \epsilon_{nmk} A_n B_m,$$

which gives the vector

$$\mathbf{A} \times \mathbf{B} = \begin{Bmatrix} A_2 B_3 - A_3 B_2, \\ A_3 B_1 - A_1 B_3, \\ A_1 B_2 - A_2 B_1 \end{Bmatrix}. \quad (30)$$

Dot products are commutative, whereas cross products are anti-symmetric:

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= +\mathbf{B} \cdot \mathbf{A}, \\ \mathbf{A} \times \mathbf{B} &= -\mathbf{B} \times \mathbf{A}. \end{aligned}$$

With Eqs. (1) and (2),

$$\begin{aligned} (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \\ &= \sum_{n=1}^3 A_n B_n \\ &\quad + \sum_{n < m} (A_n B_m - A_m B_n) \sigma_n \sigma_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3 \\ &\quad + (A_2 B_3 - A_3 B_2) i \sigma_1 \\ &\quad - (A_1 B_3 - A_3 B_1) i \sigma_2 \\ &\quad + (A_1 B_2 - A_2 B_1) i \sigma_3. \end{aligned}$$

This can be efficiently expressed in terms of the dot product and cross product as

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (31)$$

Table 6 records the same product in component form. These same dot-product and cross-product definitions extend componentwise to complex vectors in $\mathbb{C}^3$, and they apply equally to combinations of vectors, vector-valued functions, and vector operators whenever the component expressions are defined.

The reversed order flips only the antisymmetric term:

$$\begin{aligned} (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} - i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= 2\mathbf{A} \cdot \mathbf{B}, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) - (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The same product immediately gives

$$\begin{aligned} (\mathbf{A} \cdot \boldsymbol{\sigma})^2 &= \mathbf{A}^2, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{A}^2)^n, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{A}^2)^n (\mathbf{A} \cdot \boldsymbol{\sigma}), \\ (\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} &= \mathbf{A} + i\mathbf{A} \times \boldsymbol{\sigma}. \end{aligned}$$

These are the basic sigma-product identities from which the later paravector formulas are built.

Products

With Eq. (31) fixing the scalar/vector split for real vectors, the fully complex product follows as the master formula from which the simpler cases are recovered by restriction.

For Z from Eq. (15) and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$ defined analogously to Eqs. (15) and (17),

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}). \quad (32)$$

Equations (15) and (32) fix the generic complex paravector and its master multiplication rule: pure scalar, real vector, real paravector, and mixed products all follow by setting the unused parts to zero.

Products of real scalars, vectors, and paravectors

Let X and Y be the real paravectors from Eqs. (11) and (12). Using Eq. (31),

$$\begin{aligned} XY &= (a + \mathbf{A} \cdot \boldsymbol{\sigma})(b + \mathbf{B} \cdot \boldsymbol{\sigma}) \\ &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}), \\ YX &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} - i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}) \\ &= (XY)^R. \end{aligned}$$

Table 7 gives the same product in component form. The scalar part is symmetric, while the imaginary-vector part changes sign with the order.

Two immediate consequences are

$$\begin{aligned} [X, Y] &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ XY \in \mathbb{R} \oplus \mathbb{R}^3 &\iff \mathbf{A} \times \mathbf{B} = 0. \end{aligned}$$

So two real paravectors commute exactly when their vector parts are parallel.

Setting $Y = X$ removes the antisymmetric term automatically:

$$X^2 = a^2 + \mathbf{A}^2 + 2a\mathbf{A} \cdot \boldsymbol{\sigma}.$$

If $X^2 = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, then matching scalar and vector parts gives

$$\begin{aligned} b &= a^2 + \mathbf{A}^2, \\ \mathbf{B} &= 2a\mathbf{A}, \end{aligned}$$

hence

$$\begin{aligned} a^2 &= \frac{1}{2}(b \pm \sqrt{b^2 - \mathbf{B}^2}), \\ \mathbf{A}^2 &= b - a^2. \end{aligned}$$

This is the real-paravector square-root relation obtained directly from the product decomposition.

Products of complex paravectors

The master product law (32) is the complex extension of the real formulas above. Writing Z as in Eqs. (15)–(14) and $Z' = X' + iY' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, one obtains every later specialization by setting the unused scalar or vector parts to zero. Table 8 gives the full component decomposition of ZZ' , Table 9 gives Z^2 , and Tables 10–15 record the order-sensitive products built from the conjugate and reverse operators.

The scalar part of a product is cyclic:

$$\text{scalar}(ZZ') = \text{scalar}(Z'Z).$$

In particular, if $Z' = MZ^{*R}$, then

$$\begin{aligned} \text{scalar}(ZMZ^{*R}) &= \text{scalar}(MZ^{*R}Z) \\ &= \text{scalar}(M)(Z^{*R}Z), \end{aligned}$$

because $Z^{*R}Z$ is scalar.

The order-sensitive variants fall into three algebraic types:

$$\begin{aligned} ZZ^* &\text{ is a real scalar plus a complex vector,} \\ ZZ^R &\text{ is a real scalar plus a real vector,} \\ ZZ^{R*} &\text{ is a complex scalar.} \end{aligned}$$

Those three cases are precisely the ones needed later for norms, metrics, inverses, and transformation formulas.

The following tables collect the basic product decompositions and the order-dependent component expansions used throughout this subsection.

Table 6. Components of $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$.

real scalar:	$\mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$\mathbf{0}$
img vector:	$\mathbf{A} \times \mathbf{B}$

Table 7. Components of XY .

real scalar:	$ab + \mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$a\mathbf{B} + b\mathbf{A}$
img vector:	$\mathbf{A} \times \mathbf{B}$

Tables 8 and 9 collect the full component expansions for the generic product ZZ' and the special case Z^2 .

Table 8. Components of ZZ' .

real scalar:	$aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}'$
img scalar:	$ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}'$
real vector:	$a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}'$
img vector:	$a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'$

Table 9. Components of Z^2 .

real scalar:	$a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	$2ab + 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$2(a\mathbf{A} - b\mathbf{B})$
img vector:	$2(a\mathbf{B} + b\mathbf{A})$

Tables 10 and 11 make the order dependence of ZZ^* and Z^*Z explicit.

Table 10. Components of ZZ^* .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	0
real vector:	$-2\mathbf{A} \times \mathbf{B}$ $\leftarrow$ differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$

Table 11. Components of Z^*Z .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	0
real vector:	$2\mathbf{A} \times \mathbf{B}$ $\leftarrow$ differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$

Tables 12 and 13 play the same role for products involving the reverse operator.

Table 12. Components of ZZ^R .

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	0
real vector:	$2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B})$ $\leftarrow$ differs
img vector:	$\mathbf{0}$

Table 13. Components of $Z^R Z$.

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	0
real vector:	$2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B})$ $\leftarrow$ differs
img vector:	$\mathbf{0}$

Tables 14 and 15 show that the products formed with the reverse-conjugate operator collapse to purely scalar and imaginary-scalar parts.

Table 14. Components of ZZ^{R*} .

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 15. Components of $Z^{R*}Z$.

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Order dependence and commutators

The part of a product that depends on order is its commutator. For generic complex paravectors $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$,

$$[Z, Z'] := ZZ' - Z'Z = 2i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma} \quad (33)$$

Equation (33) is the compact commutator law. In real and imaginary vector components it expands to

$$[Z, Z'] = -2(\mathbf{A} \times \mathbf{B}' + \mathbf{B} \times \mathbf{A}') \cdot \boldsymbol{\sigma} + 2i(\mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}') \cdot \boldsymbol{\sigma}.$$

The real-paravector commutator

$$[X, Y] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$$

is the corresponding specialization. In every case the scalar terms cancel, so the commutator isolates the cross-product part of the multiplication law.

The same order dependence appears already in the sigma basis itself. The pseudoscalar commutes with the generators,

$$[i, \sigma_n] = 0, \quad i\boldsymbol{\sigma} = \boldsymbol{\sigma}i = \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\},$$

while distinct sigma generators satisfy

$$[\sigma_n, \sigma_m] = 2\sigma_n\sigma_m = 2 \sum_{k=1}^3 \epsilon_{nmk} i\sigma_k \quad (n \neq m).$$

Thus the oriented sigma-number-two basis is exactly the commutator closure of the sigma-number-one generators.

Tables 12–15 display the same product types in full component form. Their scalar norm and metric consequences are already fixed by Eqs. (36), (38), and (37), so no new norm or metric definitions are introduced here.

Norms and Metrics

Using Eq. (29), the product formulas above now collect the norm and scalar paravector metric identities used later in the paper. For a complex scalar $S = a + ib \in \mathbb{C}$, define the scalar norm by

$$|S|^2 := SS^* = a^2 + b^2. \quad (34)$$

If $S \in \mathbb{R}$, then Eq. (34) reduces to the ordinary absolute-value square $|S|^2 = S^2$. For a complex vector $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$, define the vector norm by

$$\|\mathbf{V}\|^2 := \mathbf{V} \cdot \mathbf{V}^* = \mathbf{A}^2 + \mathbf{B}^2. \quad (35)$$

The quadratic product $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ is not the same object as the positive norm square $\mathbf{V} \cdot \mathbf{V}^*$. For the complex paravector Z from Eqs. (15)–(14), the paravector norm is recovered from the scalar part of the product with its reverse:

$$\begin{aligned} \|Z\|^2 &:= \text{scalar}(ZZ^R) = \text{scalar}(Z^R Z) \\ &= |S|^2 + \|\mathbf{V}\|^2. \end{aligned} \quad (36)$$

This norm is therefore a real scalar. For the scalar paravector metric, use the complex paravector Z from Eqs. (15)–(14) and the real paravector X from Eq. (11). The scalar paravector metric $\langle\langle \cdot \rangle\rangle^2$ always gives a scalar: it lies in $\mathbb{C}$ for complex paravectors and in $\mathbb{R}$ for real paravectors. Define the complex scalar paravector metric by

$$\begin{aligned} \langle\langle Z \rangle\rangle^2 &:= ZZ^{*R} = Z^*R Z \\ &= S^2 - \mathbf{V}^2 \\ &= (a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2) + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \end{aligned} \quad (37)$$

The product order does not matter because $Z^{*R}Z = ZZ^{*R}$ is scalar. For the real paravector X from Eq. (11), note that $X^{*R} = X^*$, so the real scalar paravector metric is

$$\begin{aligned} \langle\langle X \rangle\rangle^2 &:= XX^{*R} = X^{*R}X = XX^* = X^*X \\ &= a^2 - \mathbf{A}^2. \end{aligned} \quad (38)$$

For $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$, the scalar paravector metric is multiplicative:

$$\begin{aligned} \langle\langle ZZ' \rangle\rangle^2 &:= ZZ'(ZZ')^{*R} = ZZ'Z'^{*R}Z^{*R} = Z(Z'Z'^{*R})Z^{*R} \\ &= Z\langle\langle Z' \rangle\rangle^2 Z^{*R} = \langle\langle Z' \rangle\rangle^2 ZZ^{*R} = \langle\langle Z \rangle\rangle^2 \langle\langle Z' \rangle\rangle^2 \\ &= \langle\langle Z'Z \rangle\rangle^2. \end{aligned} \quad (39)$$

Here $(ZZ')^{*R} = Z'^{*R}Z^{*R}$, and $\langle\langle Z \rangle\rangle^2$ and $\langle\langle Z' \rangle\rangle^2$ are scalars. Under complex scalar multiplication, the norm and scalar paravector metric scale differently:

$$\|re^{i\theta}Z\|^2 = r^2\|Z\|^2, \quad (40)$$

$$\langle\langle re^{i\theta}Z \rangle\rangle^2 = r^2e^{2i\theta}\langle\langle Z \rangle\rangle^2. \quad (41)$$

The norm is phase-insensitive, while the scalar paravector metric retains the phase square. Equations (29)–(41) therefore collect the full norm and scalar paravector metric structure used later in the paper.

Inverse formulas

The inverse formulas are the algebraic consequence of the scalar paravector metric identities from Eqs. (37) and (39). For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$, when $\langle\langle Z \rangle\rangle^2 \neq 0$, the inverse may be written in any of the equivalent forms

$$\begin{aligned} Z^{-1} &:= \frac{Z^{*R}}{\langle\langle Z \rangle\rangle^2} = \frac{Z^{*R}}{ZZ^{*R}} = \frac{Z^{*R}}{Z^*R Z} \\ &= \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2}. \end{aligned} \quad (42)$$

If the scalar paravector metric vanishes, the inverse does not exist.

Differentiating the identity $Z^{-1}Z = 1$ gives

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}. \quad (43)$$

For $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$, if $\mathbf{V} \times \mathbf{V}' = \mathbf{0}$, then

$$Z^{-1}Z' = Z'Z^{-1}. \quad (44)$$

Supporting Algebraic Derivations

The supporting derivations record the explicit sigma-basis and operator calculations, together with one generic map decomposition, that support the earlier basis and operator formulas without repeating the product development already collected above.

Basis-Element Classification

Table 1 lists the four sigma numbers and the corresponding basis-element types, while Table 2 lists the signs acquired under the conjugate operator, the reverse operator, and the reverse-conjugate operator.

Complex Paravector-Valued Maps

This notation records how real scalar and vector functions combine into a generic complex paravector-valued map without imposing any physical interpretation:

$$\begin{aligned} Z(t, \mathbf{r}) &= (a(t, \mathbf{r}) + ib(t, \mathbf{r})) \\ &\quad + (\mathbf{A}(t, \mathbf{r}) + i\mathbf{B}(t, \mathbf{r})) \cdot \boldsymbol{\sigma}, \\ a, b &: \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}, \\ \mathbf{A}, \mathbf{B} &: \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}^3, \\ Z &: \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{C} \oplus \mathbb{C}^3. \end{aligned}$$

Projector Algebra

Projectors are introduced next because they organize the later function formulas and ladder constructions. Once Eqs. (2) and (31) are fixed, the projector calculus gives the cleanest route from simple unit-direction elements to more general paravector decompositions.

If $\mathbf{A} \in \mathbb{R}^3$ is written as $\mathbf{A} = \theta \mathbf{u}$ with $\theta = \|\mathbf{A}\|$, $\mathbf{u} \in \mathbb{R}^3$, and $\mathbf{u}^2 = 1$, define the idempotents

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) \quad (45)$$

satisfying $\Pi(\pm)^2 = \Pi(\pm)$, $\Pi(+)\Pi(-) = 0$, $\Pi(+)+\Pi(-) = 1$, and $(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm)$. Algebraically, $\Pi(+)$ and $\Pi(-)$

isolate the parts aligned and anti-aligned with the direction $\mathbf{u}$. Their role here is purely structural: they split an object into two parts adapted to a chosen axis.

For a scalar-valued function $f : \{\pm 1\} \rightarrow \mathbb{C}$,

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+) + f(-)\Pi(-). \quad (46)$$

This gives a compact projector-based formula for roots, inverses, exponentials, logarithms, and related functions. Once the minimal polynomial of an STA element is known, functions of that element can be defined without leaving the algebraic framework. The detailed formulas begin with projector and eigenvalue identities and continue to ladder operations associated with a chosen axis.

About the z axis, define the ladder elements

$$\begin{aligned} a(\pm) &:= \frac{1}{2}(\sigma_1 \pm i\sigma_2), & [\sigma_3, a(\pm)] &= \pm 2a(\pm), \\ a(\pm)^2 &= 0, & a(+)a(-) &= \Pi(+), & a(-)a(+) &= \Pi(-). \end{aligned} \quad (47)$$

Equation (47) records the nilpotent combinations of σ_1 and σ_2 that exchange the two projector parts selected by Eq. (45). The projector eigenvalue formulas, mixed projector products, and ladder identities built from Eqs. (45), (46), and (47) are collected next.

Functions of a Unit-Direction Element

The projector pair $\Pi(\pm)$ gives the two-root decomposition of the unit-direction element $\mathbf{u} \cdot \boldsymbol{\sigma}$. The corresponding eigenvalue data are

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)^2 &= \Pi(\pm), \\ \Pi(+)\Pi(-) &= 0, \\ \Pi(+) + \Pi(-) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm \Pi(\pm), \end{aligned}$$

so $\mathbf{u} \cdot \boldsymbol{\sigma}$ has the two eigenvalues $+1$ and -1 . Any function with a convergent power series therefore collapses to those two scalar values:

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(+1)\Pi(+) + f(-1)\Pi(-) \\ &= \frac{1}{2}(f(+1) + f(-1)) + \frac{1}{2}(f(+1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

These are the basic projector formulas for a unit-direction element. The first examples below are chosen to illustrate the calculus before the dedicated exponential and logarithm subsections:

$$\begin{aligned} \sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \pm \Pi(+) \pm i \Pi(-), \\ (z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} &= \frac{1}{z+1}\Pi(+) + \frac{1}{z-1}\Pi(-) \\ &= \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \\ \frac{i - \mathbf{u} \cdot \boldsymbol{\sigma}}{i + \mathbf{u} \cdot \boldsymbol{\sigma}} &= i \mathbf{u} \cdot \boldsymbol{\sigma}, \end{aligned}$$

where the square root, inverse, and Cayley-type transform are obtained by direct substitution into Eq. (46). The exponential and logarithm cases are developed separately below so that the general projector method is in place first.

Functions of a Paravector

Let

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ t &= \text{scalar}(X) \in \mathbb{R}, \\ \mathbf{r} &= \text{vector}(X) \in \mathbb{R}^3. \end{aligned}$$

If $\mathbf{r} \neq 0$, write $\mathbf{r} = \|\mathbf{r}\|\mathbf{u}$ with $\mathbf{u}^2 = 1$. Then the minimal polynomial of X is

$$\begin{aligned} X^2 - 2tX + (t^2 - \mathbf{r}^2) &= 0, \\ \lambda(\pm) &= t \pm \|\mathbf{r}\|, \\ 0 &= (X - \lambda(-))(X - \lambda(+)), \end{aligned}$$

so X has the projector decomposition

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ X &= \lambda(+)\Pi(+) + \lambda(-)\Pi(-), \\ \Pi(\pm)X &= \lambda(\pm)\Pi(\pm). \end{aligned}$$

Any analytic function of X is therefore obtained by applying the scalar function to the two eigenvalues:

$$\begin{aligned} f(X) &= f(\lambda(+))\Pi(+) + f(\lambda(-))\Pi(-) \\ &= \frac{1}{2}(f(\lambda(+)) + f(\lambda(-))) \\ &\quad + \frac{1}{2}(f(\lambda(+)) - f(\lambda(-)))\mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

If $f(X) = t' + \mathbf{r}' \cdot \boldsymbol{\sigma}$, then

$$\begin{aligned} t' &= \frac{1}{2}(f(\lambda(+)) + f(\lambda(-))), \\ \|\mathbf{r}'\| &= \frac{1}{2}(f(\lambda(+)) - f(\lambda(-))), \\ \mathbf{r}' &= \|\mathbf{r}'\|\mathbf{u}. \end{aligned}$$

Before turning to exponentials and logarithms, the same construction already gives the useful square-root, inverse, and power formulas:

$$\begin{aligned} \sqrt{X} &= \pm \sqrt{\lambda(+)}\Pi(+) \pm \sqrt{\lambda(-)}\Pi(-), \\ X^{-1} &= \frac{\Pi(+)}{\lambda(+)} + \frac{\Pi(-)}{\lambda(-)} \\ &= \frac{X^*}{t^2 - \mathbf{r}^2}, \\ X^n &= \lambda(+)^n \Pi(+) + \lambda(-)^n \Pi(-). \end{aligned}$$

When two different paravectors are diagonalized along different directions, the only new ingredient is the mixed projector product. Let $X = t + \|\mathbf{r}\|\mathbf{u} \cdot \boldsymbol{\sigma}$ and $X' = t' + \|\mathbf{r}'\|\mathbf{u}' \cdot \boldsymbol{\sigma}$ be decomposed along different axes. Their functions then multiply by combining the two projector families:

$$f(X)g(X') = \sum_{s,s'=\pm} f(\lambda(s))g(\lambda'(s'))\Pi(s)\Pi'(s').$$

The mixed projector products are

$$\Pi(s)\Pi'(s') = \frac{1}{4} \left(1 + ss' \mathbf{u} \cdot \mathbf{u}' + (s\mathbf{u} + s'\mathbf{u}' + ss' i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \right),$$

with $s, s' \in \{+1, -1\}$. This compact formula reproduces all four explicit same-side and cross-side products. As a useful example, the Cayley-type transform is itself a projector exponential:

$$\begin{aligned} \frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} &= \exp(-2 \operatorname{arctanh}(\mathbf{r} \cdot \boldsymbol{\sigma})) \\ &= e^{-2 \operatorname{arctanh}(\|\mathbf{r}\|)} \Pi(+) + e^{+2 \operatorname{arctanh}(\|\mathbf{r}\|)} \Pi(-). \end{aligned}$$

Minimal Polynomial and Projector Decomposition

The two-root data of a complex paravector are collected here because the later function formulas, projector identities, and exponential identities all depend on this algebraic factorization; see Eq. (46):

$$\begin{aligned} Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0, \\ \lambda(\pm) &= S \pm \sqrt{\mathbf{V}^2}, \\ (Z - \lambda(-))(Z - \lambda(+)) &= 0. \end{aligned}$$

When $\mathbf{V}^2 \neq 0$, these two eigenvalues provide the projector data needed for the same functional calculus used in the real-paravector case.

Projector Products and Ladder Operations

The projector family in Eq. (45) and the ladder elements in Eq. (47) admit the following detailed identities:

$$\begin{aligned} \Pi(\pm)^2 &= \Pi(\pm), & \Pi(+) + \Pi(-) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm \Pi(\pm). \end{aligned}$$

with

$$\Pi(\pm)^R = \Pi(\pm), \quad \Pi(\pm)^* = \Pi(\mp).$$

For a second unit direction $\mathbf{u}'$, the same-side and cross-side products are

$$\begin{aligned} \Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm(\mathbf{u} \cdot \mathbf{u}')\Pi(\pm), \\ \Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\mp) &= \frac{1}{2}(\mathbf{u}'_{\perp} \pm i \mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma} \Pi(\mp). \end{aligned}$$

If θ is the angle between the two axes, then

$$\frac{1}{2}(1 + \mathbf{u} \cdot \mathbf{u}') = \cos^2 \frac{\theta}{2}, \quad \frac{1}{2}(1 - \mathbf{u} \cdot \mathbf{u}') = \sin^2 \frac{\theta}{2}.$$

These are the projector identities behind the standard half-angle transition factors.

The ladder elements exchange the two projector parts selected by Eq. (45). They satisfy the sharper one-sided identities

$$\begin{aligned} a(+) \Pi(+) &= 0, & a(+) \Pi(-) &= a(+), \\ \Pi(+) a(+) &= a(+), & \Pi(-) a(+) &= 0, \\ a(-) \Pi(-) &= 0, & a(-) \Pi(+) &= a(-), \\ \Pi(-) a(-) &= a(-), & \Pi(+) a(-) &= 0, \end{aligned}$$

which make the raising/lowering interpretation explicit. The original sigma generators are recovered from the same ladder/projector data:

$$\begin{aligned} \sigma_1 &= a(+) + a(-), & \sigma_2 &= \frac{a(+) - a(-)}{i}, \\ \sigma_3 &= \Pi(+) - \Pi(-). \end{aligned}$$

Commutators of Unit-Direction Elements

For unit directions $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$,

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma},$$

so the commutator is

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}.$$

Its norm is controlled by the angle between the axes,

$$\left\| \frac{1}{2} [\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] \right\|^2 = (\mathbf{u} \times \mathbf{u}')^2.$$

This is the algebraic content behind axis noncommutativity. It is an operator-level identity for the algebra elements themselves. No probabilistic or variational interpretation is assumed here.

Exponential and Logarithmic Algebra

With the projector calculus fixed, exponential and logarithmic formulas can be developed in one place. The sigma-number-one exponential produces hyperbolic half-angle transform elements, the sigma-number-two exponential produces circular half-angle transform elements, and the quaternion bridge uses that same circular basis.

Hyperbolic Exponentials and Half-Angle Elements

With the projector calculus in hand, the sigma-number-one exponential becomes especially transparent. Write $\mathbf{A} = \theta \mathbf{u}$ with $\theta = \|\mathbf{A}\|$ and $\mathbf{u}^2 = 1$. Then Eq. (46) gives

$$\begin{aligned} \exp(\pm \mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm \theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^{\pm \theta} \Pi(+) + e^{\mp \theta} \Pi(-) \\ &= \cosh(\theta) \pm \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (48)$$

With

$$\begin{aligned} \gamma &= \cosh(\theta), \\ \mathbf{V} &= \tanh(\theta) \mathbf{u}, \end{aligned}$$

Eq. (48) becomes the basic hyperbolic parametrization used repeatedly below:

$$\begin{aligned} \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 + \mathbf{V} \cdot \boldsymbol{\sigma}), \\ \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 - \mathbf{V} \cdot \boldsymbol{\sigma}). \end{aligned}$$

For the real paravector X from Eq. (11), the scalar part commutes with the vector part, so

$$\begin{aligned} \exp(X) &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}), \\ \exp(X^*) &= \exp(a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^*, \\ \exp(-X) &= \exp(-a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^{-1}. \end{aligned}$$

The corresponding half-angle element is

$$L(\theta, \mathbf{u}) = \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right), \quad X' = L^* X L. \quad (49)$$

Equation (49) is the two-sided hyperbolic transform built from Eq. (48). Its detailed transform consequences are developed in the generalized transform section. Finally, the conjugate operator simply exchanges the two projector weights:

$$\begin{aligned} \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^* &= e^{-\theta} \Pi(+) + e^{\theta} \Pi(-) \\ &= \gamma(1 - \mathbf{V} \cdot \boldsymbol{\sigma}) \\ &= \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp((\mathbf{A} \cdot \boldsymbol{\sigma})^*) \\ &= \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^{-1}. \end{aligned}$$

Logarithm of a Real Paravector

The logarithm is the inverse problem for the same projector formulas. Let Y be the real paravector from Eq. (12), and assume $b > \|\mathbf{B}\|$, so that the two real eigenvalues are positive and the principal logarithm is defined. Writing $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$ when $\mathbf{B} \neq 0$, one has

$$\begin{aligned} \lambda(\pm) &= b \pm \|\mathbf{B}\|, \\ Y &= \lambda(+) \Pi(+) + \lambda(-) \Pi(-), \end{aligned}$$

with the same projector pair $\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The logarithm therefore acts separately on the two eigenvalues:

$$\begin{aligned} \log(Y) &= \log(\lambda(+)) \Pi(+) + \log(\lambda(-)) \Pi(-) \\ &= \frac{1}{2} \log(\lambda(+)\lambda(-)) + \frac{1}{2} \log(\lambda(+)/\lambda(-)) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (50)$$

If $Y = \exp(X)$ with X from Eq. (11) and $\mathbf{A} = \alpha \mathbf{u}$, then $\lambda(\pm) = e^{a \pm \alpha}$, so the recovered parameters are

$$\begin{aligned} a &= \frac{1}{2} \log(b^2 - \mathbf{B}^2), \\ \alpha &= \operatorname{arctanh}(\|\mathbf{B}\|/b), \\ \mathbf{A} &= \alpha \mathbf{u} = \operatorname{arctanh}(\|\mathbf{B}\|/b) \frac{\mathbf{B}}{\|\mathbf{B}\|}. \end{aligned}$$

Thus the logarithm of a time-like real paravector is again a real paravector,

$$\log(Y) = a + \mathbf{A} \cdot \boldsymbol{\sigma}.$$

Circular Exponentials and Half-Angle Elements

Replacing θ by $i\theta$ turns the same projector calculation into the trigonometric form of the sigma-number-two exponential:

$$\begin{aligned} \exp(\pm i \mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm i \theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^{\pm i \theta} \Pi(+) + e^{\mp i \theta} \Pi(-) \\ &= \cos(\theta) \pm i \sin(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (51)$$

For a real angle $\theta \in \mathbb{R}$ and a unit real vector $\mathbf{u} \in \mathbb{R}^3$, the corresponding circular half-angle exponential is

$$\begin{aligned} R(\theta, \mathbf{u}) &:= \exp\left(\frac{\theta}{2} \mathbf{u} \cdot i \boldsymbol{\sigma}\right), \\ R^R R &= 1, \quad X' = R^R X R. \end{aligned} \quad (52)$$

Equation (52) is the circular transform built from Eq. (51). Its detailed transform consequences are developed in the generalized transform section. The quaternion form of the same circular exponential is recorded later in Eq. (53), after the quaternion units have been defined explicitly.

Exponentials of Complex Paravectors

Because the scalar part S commutes with $\mathbf{V} \cdot \boldsymbol{\sigma}$, the exponential again separates into scalar and directional factors:

$$\begin{aligned} X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY. \end{aligned}$$

$$\begin{aligned} \exp(Z) &= \exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \exp(S) \exp(\mathbf{V} \cdot \boldsymbol{\sigma}). \end{aligned}$$

$$\begin{aligned} \exp(Z^*) &= \exp(S^*) \exp(-\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ \exp(Z)^{-1} &= \exp(-S) \exp(-\mathbf{V} \cdot \boldsymbol{\sigma}), \\ \exp(Z^*)^{-1} &= \exp(-S^*) \exp(\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ (\mathbf{V} \cdot \boldsymbol{\sigma})^2 &= \mathbf{V} \cdot \mathbf{V} = \mathbf{V}^2, \\ &= \mathbf{A}^2 - \mathbf{B}^2 + 2i \mathbf{A} \cdot \mathbf{B}. \end{aligned}$$

The key simplification is that all higher powers collapse onto the two basis elements 1 and $\mathbf{V} \cdot \boldsymbol{\sigma}$:

$$\begin{aligned} (\mathbf{V} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{V}^2)^n, \\ z &= \sqrt{\mathbf{V}^2} \quad \text{complex scalar,} \end{aligned}$$

$$\begin{aligned} \exp(\mathbf{V} \cdot \boldsymbol{\sigma}) &= \sum_n (\mathbf{V} \cdot \boldsymbol{\sigma})^n / n!, \\ &= \sum_n (\mathbf{V}^2)^n / (2n)! + \sum_n (\mathbf{V}^2)^n / (2n+1)! (\mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \cosh(z) + \sinh(z) (\mathbf{V} \cdot \boldsymbol{\sigma}) / z. \end{aligned}$$

Quaternion Form of the Exponential Algebra

The same sigma-number-two basis used in Eq. (51) also reproduces Hamilton's quaternion units, so the exponential rotation algebra can be written in quaternion form without leaving the STA multiplication table [1, 3]:

$$\begin{aligned} q_1 &= -\sigma_2 \sigma_3, & q_2 &= -\sigma_3 \sigma_1, \\ q_3 &= -\sigma_1 \sigma_2, & \mathbf{q} &= -i \boldsymbol{\sigma}. \end{aligned} \quad (53)$$

Equation (53) fixes the identification between the quaternion units and the sigma-number-two basis elements. For $a \in \mathbb{R}$ and $\mathbf{b} \in \mathbb{R}^3$, let the quaternion $Q = a + \mathbf{b} \cdot \mathbf{q} = a - i \mathbf{b} \cdot \boldsymbol{\sigma}$. Then

$$Q Q^R = a^2 + \mathbf{b}^2, \quad Q^2 = a^2 - \mathbf{b}^2 - 2ia \mathbf{b} \cdot \boldsymbol{\sigma}. \quad (54)$$

The linear term must carry the factor a . Equation (54) makes the comparison with Eqs. (51) and (52) immediate: quaternion multiplication is the same sigma-number-two multiplication law that generates the circular exponential.

Generalized Transform Algebra

With the projector and exponential formulas fixed, the remaining purely mathematical transform material can be collected in one place. The order is generic first and special-case second: the similarity law comes first, then the basis-transport identities, then the circular and hyperbolic transforms built from the earlier exponential elements:

$$X' = T^{-1}XT. \quad (55)$$

Here T is a generic invertible algebra element. Equation (55) is the common similarity form behind the transformed-basis and ordered-product constructions that follow.

General Similarity-Transform Identities

Equation (55) gives the differential and commutator forms. In the first display, $t \in \mathbb{R}$, $\mathbf{r} \in \mathbb{R}^3$, $X = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, and T is a generic invertible algebra element:

$$\begin{aligned} X' &= T^{-1}XT, \\ &= T^{-1}(t + \mathbf{r} \cdot \boldsymbol{\sigma})T, \\ &= t + \mathbf{r} \cdot (T^{-1}\boldsymbol{\sigma}T), \\ &= t + \mathbf{r} \cdot \boldsymbol{\sigma}'. \\ T^{-1} &= T^{\mathbf{R}*}/(T^{\mathbf{R}*}T). \end{aligned}$$

This identity matters because it expresses the inverse directly in the algebra. For rotors one has $T^{\mathbf{R}*}T = 1$, so the inverse simplifies to the operator form used throughout the rotation and moving-frame calculations:

$$\begin{aligned} T^{\mathbf{R}*}T &= 1, & T^{-1} &= T^{\mathbf{R}*}, \\ T^{-1}T &= 1, & d(T^{-1})T &= -T^{-1}dT, \\ dX' &= d(T^{-1})XT + T^{-1}dXT + T^{-1}XdT, \\ &= -(T^{-1}dT)X' + T^{-1}dXT + X'(T^{-1}dT). \end{aligned}$$

Introducing the differential transform element packages the moving-frame dependence into a single algebra element. Inserting TT^{-1} into the differentiated expression then isolates the commutator contribution:

$$\begin{aligned} d\Theta &:= T^{-1}dT, \\ dX' &= T^{-1}dXT + [X', d\Theta], \\ ddX' &= d(T^{-1})dXT + T^{-1}ddXT + T^{-1}dXdT, \\ &\quad + [dX', d\Theta] + [X', dd\Theta]. \end{aligned}$$

The first derivative already shows the main structure: the carried differential $T^{-1}dXT$ is corrected only by the commutator with the moving-frame element $d\Theta$. Differentiating once more exposes the higher-order transport terms built from the same object:

$$\begin{aligned} d(T^{-1})dXT &= (-T^{-1}dT)(T^{-1}dXT), \\ &= -d\Theta(dX' - [X', d\Theta]), \\ T^{-1}dXdT &= (T^{-1}dXT)(T^{-1}dT), \\ &= (dX' - [X', d\Theta])d\Theta. \end{aligned}$$

Substituting these auxiliary identities back into ddX' yields the second-order transport law:

$$\begin{aligned} ddX' &= T^{-1}ddXT, \\ &\quad + 2[dX', d\Theta], \\ &\quad + [X', dd\Theta], \\ &\quad + [d\Theta, [X', d\Theta]]. \end{aligned}$$

This second-order formula is the clean algebraic statement behind second derivatives in a varying frame. Nothing new is added beyond the similarity transform itself; the extra pieces come only from differentiating the transforming element. For a one-parameter family $t \mapsto T(t)$, with $t \in \mathbb{R}$, the same transport equation takes the parameter-derivative form

$$\begin{aligned} d\Theta/dt &= \Omega, \\ dd\Theta/dt^2 &= d\Omega/dt, \\ dX'/dt &= T^{-1}(dX/dt)T + [X', \Omega], \\ ddX'/dt^2 &= T^{-1}(ddX/dt^2)T, \\ &\quad + 2[dX'/dt, \Omega], \\ &\quad + [X', d\Omega/dt], \\ &\quad + [\Omega, [X', \Omega]]. \end{aligned}$$

Transformed Basis Elements

The transformed basis relations below use the conjugate and reverse-operator rules of Eqs. (6) and (7):

$$\begin{aligned} T(x) &= f(x) + \mathbf{g}(x) \cdot \boldsymbol{\sigma}, \\ T^{-1} &= (f - \mathbf{g} \cdot \boldsymbol{\sigma})/(f^2 - \mathbf{g}^2), \\ dT &= df + d\mathbf{g} \cdot \boldsymbol{\sigma}, \\ \sigma'_n &:= T^{-1}\sigma_n T, \\ \sigma_n &= T\sigma'_n T^{-1}. \end{aligned}$$

Here $f(x)$ is a real scalar-valued function, $\mathbf{g}(x)$ is a real vector-valued function, and $T(x) \in \mathbb{R} \oplus \mathbb{R}^3$ is a real paravector-valued function. These definitions send the original basis into a new basis by similarity transformation. The next identities show that the transformed generators satisfy the same algebra and operator properties.

Applying the conjugate and reverse operators gives

$$\begin{aligned} (\sigma'_n)^* &= (T^{-1})^* \sigma_n^* T^*, \\ &= -(T^*)^{-1} \sigma_n T^*, \end{aligned}$$

$$\begin{aligned} (\sigma'_n)^{\mathbf{R}} &= T^{\mathbf{R}} \sigma_n^{\mathbf{R}} (T^{-1})^{\mathbf{R}}, \\ &= T^{\mathbf{R}} \sigma_n (T^{\mathbf{R}})^{-1}, \end{aligned}$$

$$\begin{aligned} (\sigma'_n)^{*R} &= -T^{*R} \sigma_n (T^{-1})^{*R}, \\ &= -T^{-1} \sigma_n T \text{ if } T^{-1} = T^{*R}, \\ &= -\sigma'_n \text{ if } T^{-1} = T^{*R}. \end{aligned}$$

Thus, whenever $T^{-1} = T^{*\mathbf{R}}$, the transformed basis inherits the same parity under the reverse-conjugate operator as the original one:

$$\begin{aligned}(\sigma'_n)^2 &= T^{-1}\sigma_n T T^{-1}\sigma_n T, \\ &= T^{-1}\sigma_n^2 T = 1.\end{aligned}$$

For $n \neq m$, the mixed products transform as follows:

$$\begin{aligned}T^{-1}\sigma_n\sigma_m T &= T^{-1}\sigma_n T T^{-1}\sigma_m T = \sigma'_n\sigma'_m, \\ &= T^{-1}\epsilon_{nmk}i\sigma_k T = \epsilon_{nmk}i\sigma'_k.\end{aligned}$$

$$\begin{aligned}\sigma'_1\sigma'_2\sigma'_3 &= T^{-1}\sigma_1 T T^{-1}\sigma_2 T T^{-1}\sigma_3 T, \\ &= T^{-1}\sigma_1\sigma_2\sigma_3 T = iT^{-1}T = i.\end{aligned}$$

So the transformed basis obeys the same sigma algebra as the original basis. The only genuinely new content appears when the transforming element depends on the parameter and derivatives of the basis must be taken. Differentiating the transformed basis makes the additional commutator terms explicit. Let $Z_s \in \mathbb{C}$ be a scalar-valued function, let $Z_{\mathbf{V}} \in \mathbb{C}^3$ be a vector-valued function, and write the corresponding complex paravector as $Z = Z_s + Z_{\mathbf{V}} \cdot \sigma'$. Then

$$\begin{aligned}d\sigma'_n &= [\sigma'_n, T^{-1}dT], \\ d\sigma' &= [\sigma', T^{-1}dT] \quad \text{vector}, \\ Z &= Z_s + Z_{\mathbf{V}} \cdot \sigma' \quad \text{paravector}, \\ \langle\langle Z \rangle\rangle^2 &= ZZ^{*\mathbf{R}}, \\ &= (Z_s + Z_{\mathbf{V}} \cdot \sigma')(Z_s - Z_{\mathbf{V}} \cdot \sigma') \text{ if } T^{-1} = T^{*\mathbf{R}}, \\ &= Z_s^2 - Z_{\mathbf{V}}^2 \text{ if } T^{-1} = T^{*\mathbf{R}}.\end{aligned}$$

$$\begin{aligned}dZ &= dZ_s + dZ_{\mathbf{V}} \cdot \sigma' + Z_{\mathbf{V}} \cdot d\sigma', \\ &= dZ_s + dZ_{\mathbf{V}} \cdot \sigma' + Z_{\mathbf{V}} \cdot [\sigma', T^{-1}dT].\end{aligned}$$

$$\begin{aligned}T^{-1}dT &= (f df - \mathbf{g} \cdot d\mathbf{g})/(f^2 - \mathbf{g}^2), \\ &+ (f d\mathbf{g} - df \mathbf{g} - i\mathbf{g} \times d\mathbf{g}) \cdot \sigma'/(f^2 - \mathbf{g}^2).\end{aligned}$$

Thus $T^{-1}dT$ contains one scalar part and one vector part that drive the transformed basis. The real vector part contributes sigma-number-one terms, while the imaginary vector part contributes sigma-number-two terms:

$$\begin{aligned}\Omega &:= T^{-1}(dT/dt), \\ \Omega' &:= T\Omega T^{-1} = (dT/dt)T^{-1}, \\ \text{scalar}(\sigma'_n) &= \text{scalar}(T^{-1}\sigma_n T), \\ &= \text{scalar}(\sigma_n), \\ &= 0.\end{aligned}$$

$$\text{scalar}(\Omega\sigma'_n) = \text{scalar}(\Omega'\sigma_n).$$

Decomposing Ω into one scalar part, one real vector part, and one imaginary vector part makes the transformed-basis decomposition explicit. Here $\Omega_s \in \mathbb{C}$, while $k = \{k_n\} \in \mathbb{R}^3$ and $\omega = \{\omega_n\} \in \mathbb{R}^3$ are real vectors:

$$\begin{aligned}k_n + i\omega_n &:= \text{scalar}(2\Omega'\sigma_n), \\ \Omega &= \Omega_s + \frac{1}{2}(k + i\omega) \cdot \sigma', \\ [\sigma', \omega \cdot (i\sigma')] &= -2\omega \times \sigma', \\ [\sigma', k \cdot \sigma'] &= 2i\mathbf{k} \times \sigma', \\ d\sigma'/dt &= [\sigma', \Omega], \\ &= i(k + i\omega) \times \sigma', \\ &= (-\omega + ik) \times \sigma'.\end{aligned}$$

Substituting this basis motion into dZ/dt produces the transport terms on the vector component:

$$\begin{aligned}Z_{\mathbf{V}} \cdot [\sigma', \Omega] &= (-Z_{\mathbf{V}} \times \omega + iZ_{\mathbf{V}} \times k) \cdot \sigma', \\ dZ/dt &= dZ_s/dt, \\ &+ (dZ_{\mathbf{V}}/dt - Z_{\mathbf{V}} \times \omega + iZ_{\mathbf{V}} \times k) \cdot \sigma'.\end{aligned}$$

Circular Directional Transforms

The identities in this subsection unpack Eq. (52) and its component consequences. Here the generator has sigma number two, so the exponential is circular rather than hyperbolic:

$$\begin{aligned}R(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot i\sigma)), \\ &= \cos(\tfrac{1}{2}\theta) + \sin(\tfrac{1}{2}\theta)(\mathbf{u} \cdot i\sigma) \quad \text{rotor.} \\ R^{-1} &= \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot i\sigma)).\end{aligned}$$

Here $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector, and R is generated by the sigma-number-two element $\mathbf{u} \cdot i\sigma$. Its inverse satisfies $R^{-1} = R^{\mathbf{R}} = R^{*\mathbf{R}}$, and the transformed paravector is

$$\begin{aligned}X' &= R^{\mathbf{R}}XR, \\ &= t' + \mathbf{r}' \cdot \sigma.\end{aligned}$$

$$\begin{aligned}t' &= t, \\ \mathbf{r}' &= \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta), \\ &+ (\mathbf{u} \cdot \mathbf{r})(1 - \cos(\theta))\mathbf{u},\end{aligned}$$

Rodrigues's formula.

Differentiating the transformed quantities gives the circular-transform analogue of the general transport law:

$$\begin{aligned}d(R^{\mathbf{R}}) &= (dR)^{\mathbf{R}} = dR^{\mathbf{R}}, \\ R^{\mathbf{R}}R &= 1, \\ dR^{\mathbf{R}}R + R^{\mathbf{R}}dR &= 0, \\ dR &= -RdR^{\mathbf{R}}R.\end{aligned}$$

The associated sigma-number-two angular-velocity element is

$$\begin{aligned}\Omega &:= -R(dR^R/ds) = (dR/ds)R^R, \\ \Omega^R &= -\Omega, \\ \Omega' &= R^R\Omega R, \\ \Omega'^R &= -\Omega', \\ dR/ds &= \Omega R.\end{aligned}$$

The resulting derivative formula has the same commutator structure as Eq. (55), with Ω constrained to be reverse-odd:

$$\begin{aligned}dX'/ds &= R^R(dX/ds)R + [X', \Omega'], \\ &= R^R(dX/ds + [X, \Omega])R.\end{aligned}$$

Action on Unit Directions

Equation (52) gives the algebraic rotation law. When the transformed object is a unit-direction element, one obtains

$$R^R(\mathbf{u} \cdot \boldsymbol{\sigma})R = \mathbf{u}' \cdot \boldsymbol{\sigma},$$

with $\mathbf{u}'$ given by the usual Rodrigues formula. If a complex paravector $\psi \in \mathbb{C} \oplus \mathbb{C}^3$ transforms as $\psi' = R\psi$, then the scalar projection built from the rotated axis obeys

$$\text{scalar}(\psi'^R(\mathbf{u} \cdot \boldsymbol{\sigma})\psi') = \text{scalar}(\psi^R(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi).$$

Here ψ is just a complex paravector. The axis information is carried entirely by the algebraic transform itself.

Hyperbolic Directional Transforms

The identities in this subsection unpack Eq. (49) and its component consequences. They are placed next to the circular identities because both come from the same exponential calculus, with sigma-number-one generators producing hyperbolic functions instead of circular ones. The auxiliary parameters β and γ are introduced first because they shorten the resulting formulas:

$$\begin{aligned}\theta &= \text{arctanh}(\beta), \\ \beta &= \tanh(\theta), \\ \gamma &= \cosh(\theta), \\ \gamma\beta &= \sinh(\theta), \\ \mathbf{u} &= (u_1, u_2, u_3), \mathbf{u}^2 = 1, \\ L(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})), \\ &= \cosh(\tfrac{1}{2}\theta) + \sinh(\tfrac{1}{2}\theta)(\mathbf{u} \cdot \boldsymbol{\sigma}). \\ L^{-1} &= \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})).\end{aligned}$$

Here $\theta, \beta, \gamma \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector, and $L \in \mathbb{R} \oplus \mathbb{R}^3$ is the hyperbolic transform element. Its inverse satisfies $L^{-1} = L^* = L^{*R}$. Elements along the same axis commute:

$$[L(\theta, \mathbf{u}), L(\theta', \mathbf{u})] = 0.$$

Consequently, collinear transforms combine by addition of the parameter θ :

$$\begin{aligned}L(\theta', \mathbf{u}) &= L(\theta_1, \mathbf{u})L(\theta_2, \mathbf{u}), \\ \beta' &= (\beta_1 + \beta_2)/(1 + \beta_1\beta_2), \\ \gamma' &= \gamma_1\gamma_2(1 + \beta_1\beta_2).\end{aligned}$$

Acting on the real paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, with $t \in \mathbb{R}$, $\mathbf{r} \in \mathbb{R}^3$, and $X \in \mathbb{R} \oplus \mathbb{R}^3$, the transform gives

$$\begin{aligned}X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ X' &= L^*XL, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\begin{aligned}t' &= t \cosh(\theta) - \sinh(\theta)(\mathbf{u} \cdot \mathbf{r}), \\ &= \gamma(t - \beta(\mathbf{u} \cdot \mathbf{r})).\end{aligned}$$

The vector part is clearest after splitting $\mathbf{r}$ into components parallel and perpendicular to the chosen direction:

$$\begin{aligned}\text{par}(\mathbf{r}) &:= (\mathbf{r} \cdot \mathbf{u})\mathbf{u} \quad \text{parallel to } \mathbf{u}, \\ \text{prp}(\mathbf{r}) &:= \mathbf{r} - \text{par}(\mathbf{r}) \quad \text{perpendicular to } \mathbf{u}, \\ \mathbf{r} &= \text{par}(\mathbf{r}) + \text{prp}(\mathbf{r}), \\ \text{par}(\mathbf{r}') &= \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ \text{prp}(\mathbf{r}') &= \text{prp}(\mathbf{r}), \\ \mathbf{r}' &= \text{par}(\mathbf{r}') + \text{prp}(\mathbf{r}'), \\ &= \text{prp}(\mathbf{r}) + \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ &= \mathbf{r} + (\gamma(\mathbf{r} \cdot \mathbf{u} - \beta t) - \mathbf{r} \cdot \mathbf{u})\mathbf{u}.\end{aligned}$$

Thus Eq. (49) gives the hyperbolic split of scalar and vector parts: the component parallel to $\mathbf{u}$ mixes with the scalar part while the perpendicular component is unchanged. In the small-parameter limit $\beta \rightarrow 0$,

$$\gamma \Rightarrow 1, t' \Rightarrow t, \mathbf{r}' \Rightarrow \mathbf{r}.$$

The real scalar paravector metric then follows directly from $L^{-1} = L^*$:

$$\begin{aligned}\langle\langle X' \rangle\rangle^2 &= X'X'^*, \\ &= (L^*XL)(L^*XL)^*, \\ &= L^*XLL^*X^*L, \\ &= L^*XX^*L, \\ &= \langle\langle X \rangle\rangle^2 L^*L, \\ &= \langle\langle X \rangle\rangle^2.\end{aligned}$$

Compositions of Hyperbolic and Circular Transforms

The ordered products are listed here with their inverses. The order matters even when each factor separately has a simple inverse rule.

For one hyperbolic element and one circular element,

$$\begin{aligned}T &= LR, \\ T^{-1} &= T^{*R} = R^{*R}L^{*R} = R^RL^*, \\ T &= RL, \\ T^{-1} &= T^{*R} = L^{*R}R^{*R} = L^*R^R.\end{aligned}$$

For two factors of the same type,

$$\begin{aligned}T &= R_1R_2, \\ T^{-1} &= T^{*R} = R_2^{*R}R_1^{*R}, \\ T &= L_1L_2, \\ T^{-1} &= T^{*R} = L_2^*L_1^*.\end{aligned}$$

These formulas are simple order identities, but they matter because they fix the inverse of each ordered product without ambiguity.

Matrix and Auxiliary Realizations

The algebra above is complete on its own, but two familiar realizations are useful to record explicitly. The standard 2×2 matrix model in Eq. (56) and determinant identity (57) give a concrete realization of the generator algebra [9], while the auxiliary 2×2 map in Eq. (58) packages a complex paravector into a diagonalizable 2×2 operator built from the same conjugate and reverse structure [12, 14]. The quaternion correspondence has already been included in the exponential and logarithmic block because it uses the same sigma-number-two basis and the same circular exponential structure as Eq. (52).

Standard 2×2 Matrix Realization

Although the algebra has been developed directly in STA form, it is useful to record a standard 2×2 matrix realization. This realization confirms the earlier product rules in a familiar concrete model without replacing the algebraic viewpoint used throughout the paper [9].

Consistent with Eq. (1), the matrix realization is

$$\sigma_1 = \{0, 1; 1, 0\}, \quad \sigma_2 = \{0, -i; i, 0\}, \quad \sigma_3 = \{1, 0; 0, -1\}, \quad (56)$$

Equation (56) gives a concrete 2×2 model for the generators. In that realization, a real paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$ becomes

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma} = \{t + r_3, r_1 - ir_2; r_1 + ir_2, t - r_3\}, \quad (57)$$

$$\det X = t^2 - \mathbf{r}^2.$$

Equation (57) reproduces exactly the real scalar paravector metric already encoded by Eq. (38). The same matrix also carries the familiar trace and characteristic data:

$$\begin{aligned} \text{tr}(X) &= 2t, \\ \lambda^2 - (\text{tr } X)\lambda + \det X &= 0, \\ \lambda(\pm) &= t \pm \|\mathbf{r}\|. \end{aligned}$$

These eigenvalues are the matrix version of the split into scalar part t and vector magnitude $\|\mathbf{r}\|$.

Within the same realization, the reverse operator and conjugate operator become concrete matrix maps. Writing

$$M^\dagger := M^{cT},$$

with M^c denoting entrywise complex conjugation, one has

$$\begin{aligned} i^c &= -i, & \sigma_1^c &= \sigma_1, & \sigma_2^c &= -\sigma_2, & \sigma_3^c &= \sigma_3, \\ (\sigma_n \sigma_m)^{cT} &= \sigma_m \sigma_n = -\sigma_n \sigma_m & (n \neq m), \\ \sigma_n^{cT} &= \sigma_n, & (i\mathbf{I})^{cT} &= -i\mathbf{I}, & (i\sigma_n)^{cT} &= -i\sigma_n, \end{aligned}$$

so in the matrix model the reverse operator coincides with the Hermitian adjoint on the sigma basis. The remaining conjugate formulas then follow from a fixed similarity matrix. With

$$\mathbf{J} := \{0, 1; -1, 0\},$$

the conjugate operator and the reverse-conjugate operator may be written as

$$M^* = \mathbf{J} M^c \mathbf{J}^{-1}, \quad M^{*R} = \mathbf{J} M^T \mathbf{J}^{-1}.$$

These identities recover the same operator rules defined abstractly in Eqs. (3)–(24).

Auxiliary 2×2 Realization

It is also useful to package a complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$ into the auxiliary 2×2 map

$$W(Z) := \{0, Z; Z^{*R}, 0\}, \quad W(Z)^2 = \text{diag}(ZZ^{*R}, Z^{*R}Z). \quad (58)$$

Here $W(Z)$ is a realization built from the same complex paravector and the reverse-conjugate operator. Its square is diagonal because the off-diagonal products collapse to the norm and scalar paravector metric formulas already introduced in Eqs. (36) and (37) [12, 14].

III. SPACE-TIME KINEMATICS

The same paravector structure also carries a space-time interpretation: the scalar paravector metric becomes the Minkowski metric of the interval, and normalized time-like paravectors become proper velocities. This interpretation follows the relativistic kinematic development of Lorentz, Einstein, and Minkowski, recast here in paravector form [4–6, 12, 13].

Differential space-time and proper velocity

Specializing the real paravector X from Eq. (11) to differentials gives

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (59)$$

and the interval is the differential version of Eq. (37):

$$ds^2 = \langle\langle dX \rangle\rangle^2 = \langle\langle dt + \boldsymbol{\sigma} \cdot d\mathbf{r} \rangle\rangle^2 = dX dX^* = dt^2 - d\mathbf{r}^2. \quad (60)$$

Here $t, s, \gamma \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and $X, U \in \mathbb{R} \oplus \mathbb{R}^3$. The scalar part dt is the temporal separation and the vector part $d\mathbf{r} \cdot \boldsymbol{\sigma}$ is the spatial separation. Equation (60) is therefore not an added metric axiom but the direct spacetime interpretation of the scalar paravector metric already introduced in Eq. (37).

If $\mathbf{v} = d\mathbf{r}/dt \in \mathbb{R}^3$ and $\gamma = (1 - \mathbf{v}^2)^{-1/2} \in \mathbb{R}$, then

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad \langle\langle U \rangle\rangle^2 = 1. \quad (61)$$

Multiplying Eq. (61) by the real rest mass $m \in \mathbb{R}$ gives the proper momentum formula

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad E = m\gamma, \quad \mathbf{P} = m\gamma\mathbf{v}, \quad (62)$$

which immediately yields $E^2 - \mathbf{P}^2 = m^2$. The normalization $\langle\langle U \rangle\rangle^2 = 1$ is the essential point: proper velocity is the unit time-like paravector tangent to the worldline. Multiplying by m then produces the energy–momentum paravector in a way that makes the relativistic mass shell relation appear immediately.

Proper acceleration and free motion

The proper acceleration is the real paravector

$$A := \frac{dU}{ds} = \frac{d^2 X}{ds^2}, \quad \text{scalar}(U^* A) = 0. \quad (63)$$

The orthogonality condition in Eq. (63) is the paravector analogue of the standard four-acceleration orthogonality relation. It is included here because the Lorentz-force law later in the paper should be read as a specific dynamical realization of this general kinematic structure.

A free-particle Lagrangian is

$$\mathcal{L}_{\text{free}} = -\frac{m}{\gamma}, \quad (64)$$

whose Euler–Lagrange equations reproduce inertial motion and the usual nonrelativistic expansion

$$\mathcal{L}_{\text{free}} = -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4). \quad (65)$$

Equation (65) is worth displaying explicitly because it shows how the relativistic action reduces to the classical kinetic expression plus an additive rest-energy constant when $\|\mathbf{v}\| \ll 1$.

Supporting Kinematic Derivations

The supporting derivations expand Eqs. (60), (61), and (62), then record the generalized-coordinate and metric remarks associated with the geometric part of the development.

Spacetime Paravectors and Interval Recovery

Starting from the displacement and interval formulas in Eqs. (59) and (60), the real paravector

$$dX = dt + \mathbf{dr} \cdot \boldsymbol{\sigma}, \quad dX^* = dt - \mathbf{dr} \cdot \boldsymbol{\sigma},$$

is the algebraic form of an infinitesimal space-time displacement. The temporal and spatial pieces are recovered directly by

$$\frac{dX + dX^*}{2} = dt, \quad \frac{dX - dX^*}{2} = \mathbf{dr} \cdot \boldsymbol{\sigma}.$$

Multiplying the displacement by its conjugate gives the interval again,

$$\langle\langle dX \rangle\rangle^2 = \langle\langle dt + \boldsymbol{\sigma} \cdot \mathbf{dr} \rangle\rangle^2 = dX dX^* = dt^2 - \mathbf{dr}^2 = ds^2.$$

This is the precise point at which the algebraic scalar paravector metric from Eq. (37) becomes the Minkowski metric of Eq. (60). For a non-null displacement, the inverse exists and is

$$(dX)^{-1} = \frac{dX^*}{ds^2}.$$

Null displacements are singled out algebraically because this inverse breaks down exactly when $ds^2 = 0$.

Interval Classes and Proper Time

Equation (60) gives the three standard interval classes without any additional geometric machinery:

$$ds^2 > 0 \Rightarrow \text{time-like},$$

$$ds^2 = 0 \Rightarrow \text{null},$$

$$ds^2 < 0 \Rightarrow \text{space-like}.$$

For a time-like worldline, the parameter s is the proper time in natural units. Writing $\mathbf{v} = d\mathbf{r}/dt$ then gives

$$ds = \sqrt{1 - \mathbf{v}^2} dt, \quad \tau = \int ds = \int \sqrt{1 - \mathbf{v}^2} dt.$$

In SI units the same relation is $ds = c d\tau$. The advantage of the paravector notation is that the interval classification and the proper-time formula both come from the single algebraic identity $dX dX^* = dt^2 - d\mathbf{r}^2$.

Proper Velocity and Proper Acceleration in Components

Starting from Eq. (61), the proper-velocity paravector can be read by its scalar and vector parts as

$$U = \frac{dX}{ds} = \gamma + \gamma \mathbf{v} \cdot \boldsymbol{\sigma},$$

$$\text{scalar}(U) = \gamma, \quad \text{vector}(U) = \gamma \mathbf{v}.$$

Its normalization $\langle\langle U \rangle\rangle^2 = 1$ implies $U^{-1} = U^*$. Differentiating with respect to s yields

$$\text{scalar}(U^* A) = 0, \quad A := \frac{dU}{ds},$$

so proper acceleration is Minkowski-orthogonal to proper velocity.

If $\mathbf{a} = d\mathbf{v}/dt$, then

$$A = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + \left(\gamma^2 \mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} \right) \cdot \boldsymbol{\sigma}.$$

Its metric is

$$\langle\langle A \rangle\rangle^2 = -\gamma^4 \left(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2 \right),$$

which is strictly space-like unless the acceleration vanishes. Splitting $\mathbf{a}$ into parts parallel and perpendicular to $\mathbf{v}$ makes the same structure more explicit:

$$\langle\langle A \rangle\rangle^2 = -\gamma^4 \left(\mathbf{a}_\perp^2 + \gamma^2 \mathbf{a}_\parallel^2 \right).$$

This is the geometric content later used by the Lorentz-force law. Tables 16, 17, and 18 record the corresponding scalar and vector parts of U , UA , and U^*A . The following table resolves the proper-velocity paravector into its scalar and spatial components.

Table 16. Components of dX/ds .

scalar:	γ
vector:	$\gamma \mathbf{v}$

The following two tables compare the products UA and U^*A , making the orthogonality structure of proper acceleration explicit.

Table 17. Components of UA .

real scalar:	$2\gamma^5(\mathbf{v} \cdot \mathbf{a})$
img scalar:	0
real vector:	$\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$
img vector:	$\gamma^3(\mathbf{v} \times \mathbf{a})$

Table 18. Components of U^*A .

real scalar:	0
img scalar:	0
real vector:	$\gamma^3\mathbf{a}$
img vector:	$-\gamma^3(\mathbf{v} \times \mathbf{a})$

Proper Momentum and the Mass Shell

Multiplying proper velocity by the rest mass produces the energy–momentum paravector,

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad E = m\gamma, \quad \mathbf{P} = m\gamma\mathbf{v}.$$

At rest one has $\mathbf{v} = 0$, hence $E = m$ in natural units and $E = mc^2$ in SI units. The kinetic energy is therefore $E - m$.

Because $\langle\langle U \rangle\rangle^2 = 1$, the mass shell follows immediately:

$$\langle\langle mU \rangle\rangle^2 = (E + \mathbf{P} \cdot \boldsymbol{\sigma})(E - \mathbf{P} \cdot \boldsymbol{\sigma}) = E^2 - \mathbf{P}^2 = m^2.$$

For $m = 0$, this reduces to $E^2 = \mathbf{P}^2$, which is the null case appropriate to massless motion.

Action Principle, Hamiltonian, and Low-Velocity Expansion

The free action is

$$S_{\text{free}} = -m \int ds = \int \mathcal{L}_{\text{free}} dt, \quad \mathcal{L}_{\text{free}} = -\frac{m}{\gamma}.$$

Differentiating with respect to $\mathbf{v}$ gives the canonical momentum again,

$$\frac{\partial \mathcal{L}_{\text{free}}}{\partial \mathbf{v}} = m\gamma\mathbf{v} = \mathbf{P}.$$

The Hamiltonian is then

$$\mathcal{H} = \mathbf{v} \cdot \frac{\partial \mathcal{L}_{\text{free}}}{\partial \mathbf{v}} - \mathcal{L}_{\text{free}} = m\gamma\mathbf{v}^2 + \frac{m}{\gamma} = m\gamma = E,$$

so the action principle reproduces both the relativistic momentum and the relativistic energy inside one calculation.

The low-velocity series

$$\begin{aligned} \mathcal{L}_{\text{free}} &= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4), \\ \mathbf{P} &= m\mathbf{v} + O(\mathbf{v}^3), \\ \mathcal{H} &= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \end{aligned}$$

shows exactly how the relativistic action contains the Newtonian kinetic term together with the additive rest-energy constant. Classical mechanics therefore appears as the approximation, not as the starting point.

Generalized Coordinate Fields and Metric Remarks

The preceding derivations depend only on the paravector algebra and the Minkowski metric, not on the particular choice of scalar coordinate t and vector coordinate $\mathbf{r}$. For a more general real paravector coordinate field,

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}, \quad dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma},$$

with $f \in \mathbb{R}$, $\mathbf{g} \in \mathbb{R}^3$, and $X \in \mathbb{R} \oplus \mathbb{R}^3$, one still has the same interval formula

$$\langle\langle dX \rangle\rangle^2 = dX dX^* = df^2 - d\mathbf{g}^2.$$

If df is time-like, the same algebraic steps as before give

$$\begin{aligned} ds &= \sqrt{df^2 - d\mathbf{g}^2}, \quad \mathbf{v} = \frac{d\mathbf{g}}{df}, \\ U &= \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad \gamma = \frac{df}{ds}. \end{aligned}$$

Thus the proper-velocity formula is a structural consequence of the paravector interval itself; changing coordinates changes only the functions f and $\mathbf{g}$, not the algebraic form of the result.

For comparison with matrix notation, one may also write

$$ds^2 = dx^T Q(x) dx$$

for a symmetric matrix $Q(x)$. Standard linear algebra then gives an orthogonal diagonalization $Q = NDN^T$. That matrix rewrite can be useful, but it does not erase the Lorentzian sign structure already built into $dXdX^*$. Diagonalization reorganizes the quadratic form; it does not convert the indefinite interval into a positive-definite Euclidean norm. The Minkowski metric therefore remains the primary statement.

IV. COVARIANT TRANSFORM APPLICATIONS

Section II established the generic similarity law (55), the circular rotor law (52), the hyperbolic boost law (49), and the ordered-product identities that accompany them. Here those formulas are applied to the interval geometry of Section III to obtain their spacetime interpretation.

When the real paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ is read as the spacetime object defined by Eqs. (59) and (60), Eq. (49) is the Lorentz transformation and the invariance of $\langle\langle X \rangle\rangle^2$ becomes the invariance of the spacetime interval. The same earlier transform formulas are then reused in Section VI for field transformations and in Section VII for ordered operator constructions, so the physical sections can refer back to one common algebraic source rather than restating it.

V. DIFFERENTIAL OPERATORS

The electromagnetic and wave-equation sections use one fixed set of differential conventions. They are collected here once so the later sections can refer back to the same numbered equations instead of redefining the operators in multiple places.

For spatial derivatives with respect to $\mathbf{r} = (r_1, r_2, r_3)$, define

$$\partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}. \quad (66)$$

The gradient notation is then

$$\nabla := \partial_{\mathbf{r}} = \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\} = \{\partial_1, \partial_2, \partial_3\}. \quad (67)$$

For shorthand, the coordinate symbol $\mathbf{r}$ is dropped in the last form, so $\partial_i := \partial/\partial r_i$ for $i \in \{1, 2, 3\}$. The corresponding spatial Laplacian is

$$\nabla^2 := \partial_1^2 + \partial_2^2 + \partial_3^2. \quad (68)$$

The STA differential paravector and its conjugate are

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \nabla = \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3. \quad (69)$$

$$\partial^* := \partial_t - \boldsymbol{\sigma} \cdot \nabla = \partial_t - \sigma_1 \partial_1 - \sigma_2 \partial_2 - \sigma_3 \partial_3. \quad (70)$$

Their product gives the d'Alembertian

$$\begin{aligned} \square &:= \langle\langle \partial \rangle\rangle^2 \\ &= \partial \partial^* \\ &= \partial^* \partial \\ &= \partial_t^2 - \nabla^2. \end{aligned} \quad (71) \quad (72)$$

VI. ELECTRO-MAGNETIC FIELDS

Electro-magnetic fields, gauge structure, and the Lorentz force admit compact STA forms. In physical content this is Maxwell theory [2], while the paravector packaging follows the later STA presentations of Hestenes, Baylis, and Doran–Lasenby [12–14].

Fields, potentials, and Maxwell equations

The differential conventions are fixed once in Section V; in particular, Eqs. (66)–(72) define the spatial derivative vector, the shorthand ∂_i , the paravector derivative, its conjugate, and the d'Alembertian. With those operators fixed, and with $V, \rho, \lambda, S, q \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{A}, \mathbf{J} \in \mathbb{R}^3$, the electromagnetic field vector and potential are written as

$$\mathbf{F} := \mathbf{E} + i\mathbf{B}, \quad F := \mathbf{F} \cdot \boldsymbol{\sigma}, \quad \Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}. \quad (73)$$

Here $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, $\mathbf{F} \in \mathbb{C}^3$ is the complex field vector, and $F \in \mathbb{R} \oplus \mathbb{C}^3$ is the corresponding STA field paravector. Equation (73) packages electric and magnetic fields into one object $\mathbf{F}$, scalar and vector potentials into one object Φ , and then lets the single symbol F carry the STA field notation used below.

Boost transformation of electro-magnetic fields

The same boost element L from Eq. (49) acts on the field paravector by

$$F' = L^* F L. \quad (74)$$

Equation (74) is the direct field analogue of the event transform $L^* X L$. It shows that F is not an extra object appended to the mechanics, but part of the same algebraic structure: boosts act on the field paravector by the same two-sided rule that acts on space-time paravectors, and the usual parallel/perpendicular split of $\mathbf{E}$ and $\mathbf{B}$ follows from the same boost algebra.

The identity

$$\partial \Phi = S + F^*, \quad S := \partial_t V + \nabla \cdot \mathbf{A} \quad (75)$$

encodes

$$\mathbf{E} = -\partial_t \mathbf{A} - \nabla V, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (76)$$

The compact Maxwell equation is

$$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (77)$$

Splitting real/imaginary scalar/vector parts of (77) reproduces

$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \cdot \mathbf{B} = 0, \quad (78)$$

$$\nabla \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \nabla \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (79)$$

Equations (78) and (79) are standard STA/GA electrodynamics up to sign conventions fixed by Eqs. (69), (70), (72), and (73); component by component they recover Maxwell's field equations in the present signature and field convention [2, 12, 13, 15]. Equation (77) avoids taking divergence and curl as separate primitive operators at this stage. They emerge automatically when the scalar and vector parts of the single STA equation are separated.

Wave equations, gauge, and Lorentz force

Applying ∂^* to (77) gives

$$\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}), \quad (80)$$

whose scalar part yields charge conservation,

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0. \quad (81)$$

Equation (81) is the continuity equation written in the present notation. For a real scalar gauge function $\lambda \in \mathbb{R}$,

$$\Phi' = \Phi + \partial^* \lambda, \quad F' = F. \quad (82)$$

This gauge transformation shows why the potential Φ is retained even when the field is physically primary: Φ carries the redundancy, while F carries the gauge-invariant derivative content.

For a particle of real mass $m \in \mathbb{R}$, real charge $q \in \mathbb{R}$, and proper velocity $U \in \mathbb{R} \oplus \mathbb{R}^3$, the proper-force formula is

$$m \frac{dU}{ds} = q \text{real}(FU) = q\gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}), \quad (83)$$

which packages both relativistic power and the Lorentz three-force. The real part in Eq. (83) extracts the physically relevant part of the product FU . The scalar term gives power transfer and the vector term gives the ordinary Lorentz force, so the standard split between energy change and momentum change is preserved inside one STA formula.

Lagrangian and energy-momentum remarks

The electromagnetic particle Lagrangian is

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad (84)$$

and its Euler–Lagrange equations return $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Here the free action from Eq. (64) is supplemented by the potential term, and the field-strength form of the force law reappears after variation.

A natural paravector candidate for field energy density and flux is

$$T := \frac{1}{2} F F^R = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (85)$$

Caution. Equation (85) is a useful frame-dependent energy-density/Poynting paravector, but by itself it is not the full electromagnetic stress-energy tensor. In STA language the full stress-energy object is more naturally a linear map such as $T(a) = \frac{1}{2} F a F^R$ [12, 14].

Supporting Electromagnetic Derivations

The supporting derivations write out the field decomposition, potential reconstruction, source formulas, gauge transformations, force-law expansions, and variational identities underlying Eqs. (66)–(85).

Electric and Magnetic Fields

This subsection expands the electromagnetic equations so that the scalar and vector content of the field paravector can be read directly. See Eqs. (66)–(72), (73), and (77). Here $V, \rho, S, q \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, $\mathbf{F} = \mathbf{E} + i\mathbf{B} \in \mathbb{C}^3$, and $F = \mathbf{F} \cdot \boldsymbol{\sigma} \in \mathbb{O} \oplus \mathbb{C}^3$. The spatial and paravector derivative notation is fixed once in Section V. In particular, Eq. (67) gives $\nabla = \{\partial_1, \partial_2, \partial_3\}$ with $\partial_i = \partial/\partial r_i$, while Eqs. (69) and (70) define ∂ and ∂^* . The operator $\partial \in \mathbb{R} \oplus \mathbb{R}^3$ is therefore a differential paravector whose scalar part is the time derivative and whose vector part is the spatial derivative. With that notation fixed, the electric and magnetic fields can be combined without introducing any new multiplication rule. Tables 19 and 20 record the scalar and vector parts of ∂Z and $i\partial Z$ for a generic complex paravector field Z . They provide the identities used repeatedly in the field expansions below.

Table 19. Components of ∂Z .

real scalar:	$\partial_t a + \nabla \cdot \mathbf{A}$
img scalar:	$\partial_t b + \nabla \cdot \mathbf{B}$
real vector:	$\nabla a + \partial_t \mathbf{A} - \nabla \times \mathbf{B}$
img vector:	$\nabla b + \nabla \times \mathbf{A} + \partial_t \mathbf{B}$

Table 20. Components of $i\partial Z$.

real scalar:	$-\partial_t b - \nabla \cdot \mathbf{B}$
img scalar:	$\partial_t a + \nabla \cdot \mathbf{A}$
real vector:	$-\nabla b - \nabla \times \mathbf{A} - \partial_t \mathbf{B}$
img vector:	$\nabla a + \partial_t \mathbf{A} - \nabla \times \mathbf{B}$

Collect the electric and magnetic fields into the complex three-vector $\mathbf{F}$ and the field paravector F :

$$\begin{aligned} \mathbf{E} &= \{E_1, E_2, E_3\} & \text{real vector electric field,} \\ \mathbf{B} &= \{B_1, B_2, B_3\} & \text{real vector magnetic field,} \\ \mathbf{F} &= \mathbf{E} + i\mathbf{B} & \text{electromagnetic field,} \\ F &= \mathbf{F} \cdot \boldsymbol{\sigma} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, & i = \sigma_1 \sigma_2 \sigma_3. \end{aligned}$$

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= -\mathbf{F}^* \cdot \boldsymbol{\sigma}. \end{aligned}$$

Define the real source paravector by

$$\begin{aligned} D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ D^* &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Here $\rho \in \mathbb{R}$ is the scalar charge density and $\mathbf{J} \in \mathbb{R}^3$ is the current-density vector. Maxwell's equations then read

$$\begin{aligned} \partial F &= D^* & \text{Maxwell's equations,} \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$\partial^* F^* = D.$$

In this form, the scalar, sigma-number-one, and sigma-number-two parts of ∂F reproduce the four classical Maxwell equations in one step. Expanding the product shows where each component sits.

Expanding ∂F into scalar, vector, and bivector parts gives

$$\begin{aligned} \partial F &= (\partial_t + \boldsymbol{\sigma} \cdot \nabla)((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}), \\ &= (\nabla \cdot \mathbf{E}), \\ &\quad + (\nabla \cdot \mathbf{B})i, \\ &\quad + (\partial_t \mathbf{E} - \nabla \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &\quad + (\partial_t \mathbf{B} + \nabla \times \mathbf{E}) \cdot (i\boldsymbol{\sigma}). \end{aligned}$$

The first two lines are the scalar and pseudoscalar parts. The last two lines are the sigma-number-one and sigma-number-two vector parts. Matching these components with $D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ immediately forces the magnetic-divergence term and Faraday term to vanish while fixing the electric source terms.

Separating the vector components then gives

$$\begin{aligned} \nabla \times \mathbf{B} &= \mathbf{J} + \partial_t \mathbf{E}, \\ \nabla \times \mathbf{E} &= -\partial_t \mathbf{B}. \end{aligned}$$

Lorentz Transformation of the Field

Once the electromagnetic field has been assembled into the single paravector F , the boost law from Eq. (74) can be expanded directly in components. For a boost $L = \exp(\frac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma})$ with $\mathbf{u}^2 = 1$, $\beta = \tanh(\theta)$, and $\gamma = \cosh(\theta)$, the parallel and perpendicular field parts relative to $\mathbf{u}$ obey

$$\begin{aligned} F' &= L^* F L, \\ \text{par}(\mathbf{E}') &= \text{par}(\mathbf{E}), \\ \text{par}(\mathbf{B}') &= \text{par}(\mathbf{B}), \\ \text{prp}(\mathbf{E}') &= \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \text{prp}(\mathbf{B}') &= \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}). \end{aligned}$$

Equivalently, the transformed fields can be written in the compact form

$$\begin{aligned} \mathbf{E}' &= \text{par}(\mathbf{E}) + \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \mathbf{B}' &= \text{par}(\mathbf{B}) + \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}), \\ \partial' &= L^* \partial L, \\ \partial' F' &= L^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})L. \end{aligned}$$

The parallel components are unchanged, while the perpendicular components mix by the standard electric-magnetic cross terms. This is the component form of the single algebraic boost law $F' = L^* F L$ already introduced in Eq. (74).

Wave Operator and d'Alembertian

The wave-operator conventions are fixed once in Eqs. (69), (70), and (72). Only the derived identities needed later are recorded here:

$$\partial^2 = \partial_t^2 + \nabla^2 + 2\partial_t \nabla \cdot \sigma.$$

Equation (72) should be distinguished from ∂^2 . The present convention follows Eqs. (73) and (77); different sources may reverse the sign of F , adopt the metric signature $(-, +, +, +)$, or use a different box notation. The corresponding wave equation is

$$\square F = \partial^* \partial F = \partial^* (\rho - \mathbf{J} \cdot \sigma).$$

$$\begin{aligned} \square F^* &= \partial(\rho + \mathbf{J} \cdot \sigma), \\ &= (\square \mathbf{E} + i \square \mathbf{B}) \cdot \sigma, \\ &= (\partial_t \rho + \nabla \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \nabla \rho + i(\nabla \times \mathbf{J})) \cdot \sigma. \end{aligned}$$

$$\partial_t \rho = -\nabla \cdot \mathbf{J} \quad \text{conservation of charge.}$$

The scalar part of $\square F^*$ is therefore the continuity equation, while Eq. (80) and the vector part of $\square F^*$ give the sourced wave equations for $\mathbf{E}$ and $\mathbf{B}$. In vacuum both source terms disappear and each field component satisfies the ordinary wave equation.

In vacuum, where $\rho = 0$ and $\mathbf{J} = 0$, this reduces to

$$\begin{aligned} \square \mathbf{E} &= 0 \Rightarrow \partial_t^2 \mathbf{E} = \nabla^2 \mathbf{E}, \\ \square \mathbf{B} &= 0 \Rightarrow \partial_t^2 \mathbf{B} = \nabla^2 \mathbf{B}. \end{aligned}$$

Tables 21 and 22 record the scalar and vector parts of Maxwell's equation and its sourced wave consequence.

Table 21. Components of $\partial F = \rho - \mathbf{J} \cdot \sigma$.

real scalar:	$\nabla \cdot \mathbf{E} = \rho$
img scalar:	$\nabla \cdot \mathbf{B} = 0$
real vector:	$\partial_t \mathbf{E} - \nabla \times \mathbf{B} = -\mathbf{J}$
img vector:	$\partial_t \mathbf{B} + \nabla \times \mathbf{E} = 0$

Table 22. Components of $\square F = \partial^* (\rho - \mathbf{J} \cdot \sigma)$.

real scalar:	$\partial_t \rho + \nabla \cdot \mathbf{J} = 0$
img scalar:	$0 = 0$
real vector:	$\square \mathbf{E} = -\partial_t \mathbf{J} - \nabla \rho$
img vector:	$\square \mathbf{B} = \nabla \times \mathbf{J}$

Potential Fields

Eqs. (75) and (76) put gauge freedom, source reconstruction, and the electric/magnetic field split in one place. Here $V, S, \rho, \lambda \in \mathbb{R}$,

$\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, $\mathbf{F} \in \mathbb{C}^3$, and $F \in \mathbb{C} \oplus \mathbb{C}^3$:

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \sigma, \\ \Phi &= V + \mathbf{A} \cdot \sigma, \\ \partial \Phi &= \partial_t V + \nabla \cdot \mathbf{A}, \\ &\quad + (\partial_t \mathbf{A} + \nabla V) \cdot \sigma, \\ &\quad + i(\nabla \times \mathbf{A}) \cdot \sigma. \end{aligned}$$

This expansion separates the potential gradient into one real scalar part and one pure-vector field part. The scalar piece becomes the Lorenz-gauge scalar S , and the remaining vector piece becomes the electromagnetic field:

$$\begin{aligned} S &:= \text{scalar}(\partial \Phi), \\ &= \partial_t V + \nabla \cdot \mathbf{A}. \end{aligned}$$

The component identities extracted from $\partial \Phi$ are

$$\begin{aligned} \mathbf{E} &= -\partial_t \mathbf{A} - \nabla V, \\ \mathbf{B} &= \nabla \times \mathbf{A}. \end{aligned}$$

The identity is explicit:

$$\begin{aligned} \partial \Phi &= S + F^*, \\ F^* &= \partial \Phi - S, \\ \text{scalar}(F^*) &= \text{scalar}(\partial \Phi - S) = 0, \\ \text{vector}(F^*) &= \text{vector}(\partial \Phi). \end{aligned}$$

The field is therefore the non-scalar part of $\partial \Phi$. Written this way, gauge freedom changes only the scalar part S and leaves the field paravector itself unchanged. Table 23 records the component content of the potential identity $\partial \Phi = S + F^*$.

Table 23. Components of $\partial \Phi = S + F^*$.

real scalar:	$\partial_t V + \nabla \cdot \mathbf{A} = S$
img scalar:	$0 = 0$
real vector:	$\partial_t \mathbf{A} + \nabla V = -\mathbf{E}$
img vector:	$\nabla \times \mathbf{A} = \mathbf{B}$

The differential operator acts before the conjugate and reverse operators are applied, so

$$(\partial \Phi)^{*R} \neq \Phi^{*R} \partial^{*R}.$$

Equivalently, without introducing S , one may write

$$F^* = (\partial \Phi - (\partial \Phi)^{*R})/2.$$

Sources Derived from the Potential

Using the previously defined potential $\Phi = V + \mathbf{A} \cdot \sigma$, scalar part $S = \partial_t V + \nabla \cdot \mathbf{A}$, and identity $F^* = \partial \Phi - S$, one obtains

$$\begin{aligned} F &= \partial^* \Phi^* - S, \\ \partial F &= \partial \partial^* \Phi^* - \partial S, \\ &= \square \Phi^* - \partial S, \\ &= (\square V - \partial_t S) + (-\square \mathbf{A} - \nabla S) \cdot \sigma, \\ &= \rho - \mathbf{J} \cdot \sigma. \end{aligned}$$

$$\begin{aligned}\rho &= \square V - \partial_t S, \\ \mathbf{J} &= \square \mathbf{A} + \nabla S.\end{aligned}$$

These are the component source-reconstruction formulas hidden inside the compact Maxwell equation. The scalar equation determines charge density from the potential data, and the vector equation determines current density from the same data.

In Lorenz gauge, where $S = 0$, these relations reduce to

$$\begin{aligned}\partial_t V &= -\nabla \cdot \mathbf{A}, \\ \rho &= \square V, \\ \mathbf{J} &= \square \mathbf{A}.\end{aligned}$$

Gauge Transform of the Potential

$$\begin{aligned}\text{scalar}(\partial\Phi) &= S, \\ S &= \partial_t V + \nabla \cdot \mathbf{A}.\end{aligned}$$

Let λ be a scalar gauge function. Then

$$\begin{aligned}\Phi' &= \Phi + \partial^* \lambda, \\ \partial\Phi' &= \partial(\Phi + \partial^* \lambda), \\ &= \partial\Phi + \partial(\partial^* \lambda), \\ &= \partial\Phi + \square \lambda.\end{aligned}$$

Since $\square \lambda$ is again a scalar field, only the scalar part changes:

$$\begin{aligned}\text{scalar}(\partial\Phi') &= S + \square \lambda, \\ \text{vector}(\partial\Phi') &= \text{vector}(\partial\Phi), \\ F' &= F.\end{aligned}$$

This makes the gauge split explicit: Φ changes by a derivative of a real scalar, the scalar part of $\partial\Phi$ shifts by $\square \lambda$, and the field paravector remains unchanged.

Proper Force and Lorentz Dynamics

$$\begin{aligned}mU &= mdX/ds, \\ &= m\gamma + m\gamma \mathbf{v} \cdot \boldsymbol{\sigma}, \\ d/ds &= \gamma d/dt.\end{aligned}$$

The scalar part of $d(mU)/dt$ is the power equation and the vector part is the three-force equation. Writing both pieces in one paravector equation makes the usual split visible without abandoning the relativistic form.

The time component gives the relativistic work law,

$$d/dt(m\gamma) = q\mathbf{v} \cdot \mathbf{E}.$$

Here $\mathbf{E}$ denotes the electric field, not the energy operator E , and the magnetic field contributes no work term in the scalar component. The spatial component gives the relativistic Lorentz force law,

$$\begin{aligned}\mathbf{P} &:= m\gamma \mathbf{v}, \\ d\mathbf{P}/dt &= q\mathbf{E} + q\mathbf{v} \times \mathbf{B}.\end{aligned}$$

Together these assemble into

$$d/dt(mU) = q\mathbf{v} \cdot \mathbf{E} + (q\mathbf{E} + q\mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

To expose the paravector content explicitly, expand

$$\begin{aligned}FU/\gamma &= ((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + (\mathbf{E} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &\quad + i(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + \mathbf{E} \cdot \mathbf{v}, \\ &\quad + i(\mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \mathbf{v} - (\mathbf{B} \times \mathbf{v}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The product FU naturally separates into real and imaginary parts. The real part reproduces the physical Lorentz-force content, while the imaginary part exchanges electric and magnetic roles and is therefore not used in the force law itself. Table 24 gives the component form of the product FU/γ used in the Lorentz-force law.

Table 24. Components of FU/γ .

real scalar:	$\mathbf{E} \cdot \mathbf{v}$
img scalar:	$\mathbf{B} \cdot \mathbf{v}$
real vector:	$\mathbf{E} + \mathbf{v} \times \mathbf{B}$
img vector:	$\mathbf{B} + \mathbf{E} \times \mathbf{v}$

$$\begin{aligned}\text{real}(FU) &= \frac{1}{2}(FU + UF^R), \\ &= \gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}).\end{aligned}$$

The imaginary part of FU exchanges the electric and magnetic roles and is therefore not the part used in the physical force law:

$$\begin{aligned}mdU/ds &= q\text{real}(FU), \\ &= \frac{1}{2}q(FU + UF^R).\end{aligned}$$

$$mddX/ds^2 = q\text{real}(FdX/ds) \quad \text{4-force.}$$

Variational Derivation of the Lorentz Force

Starting from the Lagrangian of Eq. (84), together with the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, proper velocity $U = dX/ds$, and $\gamma = dt/ds$, the action takes the form

$$\begin{aligned}\text{action} &= -m \int ds - q \int \text{scalar}(\Phi^* U) ds, \\ &= \int (-m/\gamma - q\text{scalar}(\Phi^* U/\gamma)) dt, \\ &= \int (-m/\gamma, \\ &\quad - q\text{scalar}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}))) dt, \\ &= \int (-m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}) dt.\end{aligned}$$

$$\mathcal{L} = -m/\gamma - q(V - \mathbf{v} \cdot \mathbf{A}) \quad \text{Lagrangian.}$$

This is the standard minimal-coupling Lagrangian written in the present paravector notation. The canonical momentum is shifted

by the vector potential, exactly as in the ordinary component derivation.

The canonical momentum is

$$\begin{aligned}\partial_{\mathbf{v}}\mathcal{L} &= m\gamma\mathbf{v} + q\mathbf{A}, \\ &= \mathbf{P} + q\mathbf{A}.\end{aligned}$$

$$\begin{aligned}d/dt\partial_{\mathbf{v}}\mathcal{L} &= d\mathbf{P}/dt + q\partial_t\mathbf{A} + q(\mathbf{v} \cdot \nabla)\mathbf{A}, \\ \nabla\mathcal{L} &= -q\nabla V + q\nabla(\mathbf{v} \cdot \mathbf{A}).\end{aligned}$$

Using the vector-calculus identity

$$(\mathbf{v}(t) \cdot \nabla)\mathbf{A}(t, \mathbf{r}) = \nabla(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\nabla \times \mathbf{A}).$$

the Euler–Lagrange equation $d/dt\partial_{\mathbf{v}}\mathcal{L} = \nabla\mathcal{L}$ yields

$$\begin{aligned}d\mathbf{P}/dt &= -q\nabla V - q\partial_t\mathbf{A}, \\ &+ q(\nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A}), \\ &= q(-\nabla V - \partial_t\mathbf{A} + \mathbf{v} \times (\nabla \times \mathbf{A})), \\ &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \\ &= m\text{vector}(dU/ds)/\gamma.\end{aligned}$$

$$\begin{aligned}\mathcal{H} &= (\partial_{\mathbf{v}}\mathcal{L}) \cdot \mathbf{v} - \mathcal{L}, \\ &= m\gamma\mathbf{v} \cdot \mathbf{v} + q\mathbf{v} \cdot \mathbf{A} + m/\gamma.\end{aligned}$$

Adding the explicit potential contribution $qV - q\mathbf{v} \cdot \mathbf{A}$ gives

$$\begin{aligned}&= m\gamma(\mathbf{v}^2 + 1/\gamma^2) + qV, \\ &= m\gamma(\mathbf{v}^2 + 1 - \mathbf{v}^2) + qV, \\ &= m\gamma + qV.\end{aligned}$$

The resulting Hamiltonian is the relativistic energy plus the scalar potential term. Using the Hamiltonian identity $dH/dt = -\partial_t\mathcal{L}$,

$$\begin{aligned}dH/dt &= -\partial_t\mathcal{L}, \\ d\mathcal{H}/dt &= d(m\gamma)/dt + qdV/dt, \\ &= d(m\gamma)/dt + q\partial_t V + q\mathbf{v} \cdot \nabla V.\end{aligned}$$

$$\begin{aligned}\partial_t H &= q\partial_t V - q\mathbf{v} \cdot \partial_t\mathbf{A}, \\ d(m\gamma)/dt &= q\mathbf{v} \cdot (-\nabla V - \partial_t\mathbf{A}), \\ &= q\mathbf{v} \cdot \mathbf{E}.\end{aligned}$$

Electromagnetic Energy-Momentum

The next identities collect the simplest field invariants built from F and the frame-dependent paravector $T = \frac{1}{2}FF^R$. They are useful reference formulas, but the caution in the main text remains in force: T is not by itself the full stress-energy tensor:

$$\begin{aligned}F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ F^R &= (\mathbf{E} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ T &= \frac{1}{2}FF^R, \\ &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= T_s + T_{\mathbf{v}} \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \nabla.$$

This is a frame-dependent energy-density/Poynting paravector, not by itself the full electromagnetic stress-energy tensor.

Using the generic paravector product rule,

$$\begin{aligned}\partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \nabla Z', \\ \partial T &= \frac{1}{2}\partial(FF^R), \\ &= \frac{1}{2}(\partial F)F^R + \frac{1}{2}F(\partial F^R) \\ &\quad + \frac{1}{2}[\boldsymbol{\sigma}, F] \cdot \nabla F^R.\end{aligned}$$

$$D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \quad \text{density paravector,}$$

$$D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma},$$

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \nabla,$$

$$\partial F = D^*,$$

$$F^2 = (\mathbf{E}^2 - \mathbf{B}^2) + 2i(\mathbf{E} \cdot \mathbf{B}) \quad \text{scalar.}$$

Applying the same product rule to F^2 gives

$$\begin{aligned}\partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \nabla Z', \\ \partial(F^2) &= (\partial F)F + F(\partial F) \\ &\quad + [\boldsymbol{\sigma}, F] \cdot \nabla F, \\ &= D^*F + FD^* + [\boldsymbol{\sigma}, F] \cdot \nabla F.\end{aligned}$$

$$[\boldsymbol{\sigma}, F] = 2i\mathbf{F} \times \boldsymbol{\sigma},$$

$$\begin{aligned}[\boldsymbol{\sigma}, F] \cdot \nabla F &= -2i\mathbf{F} \cdot (\nabla \times \mathbf{F}), \\ &\quad - 2((\nabla \cdot \mathbf{F})\mathbf{F} - \frac{1}{2}\nabla F^2) \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\square F^2 = \partial^* \partial(F^2) \quad \text{scalar.}$$

These formulas isolate the two standard electromagnetic invariants, $\mathbf{E}^2 - \mathbf{B}^2$ and $\mathbf{E} \cdot \mathbf{B}$, and show how their derivatives are packaged algebraically. They support these invariant identities, while the full stress-energy discussion remains the one stated around Eq. (85). Tables 25 and 26 summarize the two simplest invariants built from F and the frame-dependent energy-density/Poynting paravector $T = \frac{1}{2}FF^R$.

Table 25. Components of $\frac{1}{2}F^2$.

real scalar:	$\frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$i\mathbf{E} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 26. Components of $\frac{1}{2}FF^R$.

scalar:	$\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$	energy density
vector:	$\mathbf{E} \times \mathbf{B}$	Poynting vector

The remaining tables record the differentiated and source-coupled field combinations used in the energy-momentum discussion.

Table 27. Components of ∂T .

real scalar:	$\partial_t T_s + \nabla \cdot T_v$
img scalar:	0
real vector:	$\nabla T_s + \partial_t T_v$
img vector:	$\nabla \times T_v$

Table 28. Components of ∂F^2 .

real scalar:	$\partial_t(\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$\partial_t(2\mathbf{E} \cdot \mathbf{B})$
real vector:	$\nabla(\mathbf{E}^2 - \mathbf{B}^2)$
img vector:	$\nabla(2\mathbf{E} \cdot \mathbf{B})$

Table 29. Components of FD^* .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho\mathbf{E} + \mathbf{B} \times \mathbf{J}$
img vector:	$\rho\mathbf{B} - \mathbf{E} \times \mathbf{J}$

Table 30. Components of D^*F .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho\mathbf{E} - \mathbf{B} \times \mathbf{J}$
img vector:	$\rho\mathbf{B} + \mathbf{E} \times \mathbf{J}$

Table 31. Components of $D^*F + FD^*$.

real scalar:	$-2\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-2\mathbf{B} \cdot \mathbf{J}$
real vector:	$2\rho\mathbf{E}$
img vector:	$2\rho\mathbf{B}$

VII. RELATIVISTIC WAVE EQUATIONS AND SPIN DYNAMICS

The relativistic wave-equation development is a second application of the same algebra: first the scalar wave equation, then minimal coupling and factorization, and only then the two-component paravector factorization, the projector split, and the Pauli limit. The sequence follows the historical progression from Klein and Gordon to Dirac and Pauli, but it is rewritten throughout in the common paravector language established earlier [7–10].

Klein–Gordon and frequency splitting

For a complex scalar field $\Psi \in \mathbb{C}$,

$$(\partial\partial^* + m^2)\Psi = 0 \quad (86)$$

is the free Klein–Gordon equation [7, 8]. Define also

$$\Omega := \langle\langle m + \boldsymbol{\sigma} \cdot \nabla \rangle\rangle = \sqrt{m^2 - \nabla^2}, \quad (87)$$

$$(\partial_t \pm i\Omega)\Psi(\pm) = 0, \quad \Psi = \Psi(+) + \Psi(-),$$

which is the positive/negative-frequency factorization of (86). This factorization is useful both conceptually and technically. It

separates opposite-frequency branches of the scalar theory, and it also prepares the reader for the later factorizations in which second-order relativistic equations are rewritten as first-order systems.

Minimal coupling and operator factorization

Minimal coupling enters directly at the operator level. Rather than modifying each component equation separately, one defines energy and momentum operators that already incorporate the potential Φ established in Section VI.

Here $\Psi(\pm) \in \mathbb{C}$, while Ω is a scalar differential operator built from $m \in \mathbb{R}$ and ∇ . Define also the scalar differential operator E and vector differential operator $\mathbf{P}$ by

$$E := i\partial_t - qV, \quad \mathbf{P} := -i\nabla - q\mathbf{A}. \quad (88)$$

Then

$$(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0 \quad (89)$$

expands to the minimal-coupling Klein–Gordon operator

$$\left(\square + iqS - q^2\langle\langle\Phi\rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \nabla)\right)\Psi = 0. \quad (90)$$

The basic covariant-momentum commutators are

$$[E, P_m] = iqE_m, \quad [P_m, P_n] = iq(\partial_m A_n - \partial_n A_m), \quad (91)$$

and using Eq. (91) one gets the useful factorization

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF. \quad (92)$$

Equation (92) is the bridge between the Klein–Gordon and Dirac-type formulas below. The square of a first-order operator built from E and $\mathbf{P}$ produces the second-order relativistic operator together with the field term, which is the structural fact needed before introducing a Dirac-type equation.

Dirac Two-Component Form, Gauge Covariance, and Pauli Limit

Using the auxiliary 2×2 map from Eq. (58), the free and coupled Dirac operators act on a two-component field $\Psi = \{\Psi_1; \Psi_2\}$ whose components $\Psi_1, \Psi_2 \in \mathbb{C} \oplus \mathbb{C}^3$ are complex paravectors:

$$D_0 := W(i\partial) - m\mathbf{I}, \quad D_0\Psi = 0, \quad (93)$$

$$D := W(i\partial - q\Phi^*) - m\mathbf{I}, \quad D\Psi = 0. \quad (94)$$

Multiplying (94) by $D' := W(i\partial - q\Phi^*) + m\mathbf{I}$ yields

$$D'D\Psi = 0 \Rightarrow \begin{cases} (-E^2 + \mathbf{P}^2 + m^2 - iqF^*)\Psi_1 = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2 = 0. \end{cases} \quad (95)$$

The calculation in Eq. (95) explains why the two-component construction is more than formal packaging. Squaring the first-order system returns Klein–Gordon-type equations with the expected field couplings, so the Dirac form is seen as a controlled factorization of the relativistic wave operator. For the ordering $Z = i\partial - q\Phi^* = E - \boldsymbol{\sigma} \cdot \mathbf{P}$ inside $W(Z)$, the upper component carries $-iqF^*$ and the lower component carries $+iqF$.

Gauge covariance combines the electromagnetic gauge law (82) with the phase transformation of that same two-component field,

$$\Psi = e^{iq\lambda} \Psi', \quad (96)$$

Equation (96) leaves the form of (94) unchanged.

For the low-energy reduction, using the operators from Eq. (88), one obtains the leading-order Pauli equation for the same two-component field Ψ ,

$$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi. \quad (97)$$

Solving Eq. (97) for $i\partial_t\Psi$ gives the more familiar Schrödinger-type form

$$i\partial_t\Psi \approx \left(\frac{(-i\nabla - q\mathbf{A})^2}{2m} + qV \right) \Psi - \frac{q}{2m} (\mathbf{B} \cdot \boldsymbol{\sigma})\Psi. \quad (98)$$

This matches the usual Zeeman term [9, 17], provided one remembers that (98) is a leading-order $1/m$ approximation and that later specialization to uniform static $\mathbf{B}$ is used when decoupling the spin-up and spin-down parts. The nonrelativistic theory therefore appears as an approximation internal to the relativistic one, not as an unrelated starting point. The Pauli equation is the leading low-energy shadow of the Dirac two-component field once the rest-energy scale has been separated and higher-order corrections are neglected.

If $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$ with $\mathbf{B}, \mathbf{u} \in \mathbb{R}^3$, $\|\mathbf{u}\| = 1$, and $\mathbf{B}$ uniform and constant, decompose

$$\begin{aligned} \Psi &= \Psi(+) + \Psi(-), & \Psi(\pm) &= \Pi(\pm)\Psi, \\ \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (99)$$

With the projector decomposition (99) in place, Eq. (98) separates into

$$i\partial_t\Psi(\pm) = ((\mathbf{P}^2 \mp q\|\mathbf{B}\|)/(2m) + qV) \Psi(\pm). \quad (100)$$

In this special case the projector language acquires an immediate physical meaning: the field direction selects a preferred axis, and the two projector parts evolve with opposite Zeeman shifts. If the field direction or magnitude varies, the same projector family from Section II still organizes the state but need not decouple the dynamics. Writing

$$H := \frac{\mathbf{P}^2}{2m} + qV - \frac{q\|\mathbf{B}\|}{2m} \mathbf{u} \cdot \boldsymbol{\sigma}, \quad i\partial_t\Psi = H\Psi,$$

the projector-selected parts obey

$$\begin{aligned} i\partial_t\Psi(\pm) &= i(\partial_t\Pi(\pm))\Psi + \Pi(\pm)(i\partial_t\Psi) \\ &= H\Psi(\pm) + S(\pm), \end{aligned}$$

with coupling term

$$S(\pm) := (i\partial_t\Pi(\pm) + [\Pi(\pm), H])\Psi.$$

The uniform static-field split in Eq. (100) is precisely the case $S(\pm) = 0$.

Energy Projectors and On-Shell Split

The projector and ladder identities from Section II are purely algebraic. For the relativistic two-component field, the additional physics-specific split comes from the unit time-like paravector U in Eq. (61) and the auxiliary 2×2 map W in Eq. (58). Define for $\epsilon, s \in \{+, -\}$

$$\Lambda(\pm) := \frac{1}{2}(\mathbf{I} \pm W(U)), \quad \Psi(\epsilon, s) := \Lambda(\epsilon)\Pi(s)\Psi. \quad (101)$$

Here $\Lambda(\pm)$ are the energy projectors built from the unit time-like paravector U , while $\Pi(\pm)$ are the axis projectors already defined in Eq. (45). Equation (101) therefore combines the time-like split from Eq. (61) with the earlier algebraic axis split. Since Eq. (61) gives $\langle\langle U \rangle\rangle^2 = 1$, one has $W(U)^2 = \mathbf{I}$, hence $\Lambda(\pm)^2 = \Lambda(\pm)$ and

$$\Psi = \Psi(+, +) + \Psi(+, -) + \Psi(-, +) + \Psi(-, -).$$

After U has been normalized and the free state has been restricted to the positive-energy branch, one may also write the on-shell projector condition

$$(W(U) - \mathbf{I})\Psi = 0. \quad (102)$$

Equation (102) is not a replacement for the Dirac equation; it is an on-shell projector relation for a free state satisfying $W(U)^2 = \mathbf{I}$ after momentum has been normalized by m .

For a normalized complex paravector state $\psi \in \mathbb{C} \oplus \mathbb{C}^3$, write the scalar projections

$$\begin{aligned} \langle\psi | \psi\rangle &:= \text{scalar}(\psi^R \psi) = 1, \\ \langle\psi | M | \psi\rangle &:= \text{scalar}(\psi^R M \psi). \end{aligned}$$

Here M may be any algebra element with a well-defined product on ψ . The bracket notation is only a compact mnemonic for scalar extraction on normalized paravector states; the underlying objects remain paravectors throughout.

Supporting Wave-Equation Derivations

The supporting derivations expand the scalar and two-component paravector wave equations into their intermediate operator identities, explicit component forms, Fourier interpretations, low-energy limits, and spin decompositions.

Spin-0 Wave Mechanics and Frequency Splitting

Equation (86) is most transparent when read one Fourier mode at a time. For a plane-wave ansatz

$$\Psi(t, \mathbf{r}) = c e^{-i\omega t + i\mathbf{k} \cdot \mathbf{r}},$$

the Klein–Gordon equation becomes $\omega^2 = \mathbf{k}^2 + m^2$. The operator Ω in Eq. (87) therefore acts on each Fourier mode by multiplication with the positive root

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

Accordingly, the symbol Ω in Eq. (87) should be read as the positive scalar square root of the operator $m^2 - \nabla^2$. The factorization

is therefore defined one Fourier mode at a time rather than treated as shorthand only.

Equivalently,

$$\partial_t^2 + \Omega^2 = (\partial_t - i\Omega)(\partial_t + i\Omega).$$

The decomposition $\Psi = \Psi(+) + \Psi(-)$ is thus not merely notation: it separates the positive- and negative-frequency branches of the scalar theory in the same way that the Fourier transform separates the two energy branches [16].

Root Operators and Fourier Interpretation

The square-root operators in Eq. (87) are easiest to interpret in Fourier space. If

$$\Psi = \sum_n c_n e^{i(\omega_n t + \mathbf{k}_n \cdot \mathbf{r})},$$

then

$$\begin{aligned} \square \Psi &= \sum_n (\mathbf{k}_n^2 - \omega_n^2) c_n e^{i(\omega_n t + \mathbf{k}_n \cdot \mathbf{r})}, \\ \Omega \Psi &= \sum_n \sqrt{m^2 + \mathbf{k}_n^2} c_n e^{i(\omega_n t + \mathbf{k}_n \cdot \mathbf{r})}. \end{aligned}$$

The factorization

$$\square + m^2 = (m - i\sqrt{\square})(m + i\sqrt{\square})$$

is therefore a shorthand for multiplying each Fourier mode by the corresponding dispersion function. The apparently formal root operator is nothing more than the same function rule already introduced in Eq. (46), now applied to the wave operator.

Minimal Coupling and Operator Factorization

Once the potential Φ has been introduced, it is more economical to couple at the operator level than at the component level. Acting on a smooth test field Ψ , the operators from Eq. (88) satisfy

$$[E, P_m]\Psi = iqE_m\Psi, \quad [P_m, P_n]\Psi = iq(\partial_m A_n - \partial_n A_m)\Psi.$$

These commutators are the exact place where the electromagnetic field enters the coupled wave mechanics: electric components arise from time-space commutators, while magnetic components arise from the noncommutativity of spatial covariant momenta.

Expanding the squared operators themselves gives

$$\begin{aligned} E^2 \Psi &= (-\partial_t^2 - iq\partial_t V - 2iqV\partial_t + q^2 V^2) \Psi, \\ \mathbf{P}^2 \Psi &= (-\nabla^2 + iq\nabla \cdot \mathbf{A} + 2iq\mathbf{A} \cdot \nabla + q^2 \mathbf{A}^2) \Psi. \end{aligned}$$

Subtracting these expressions and then adding $m^2\Psi$ is the direct operator expansion behind Eq. (90).

Substituting those commutators into the product of first-order factors yields

$$(E + \sigma \cdot \mathbf{P})(E - \sigma \cdot \mathbf{P})\Psi = (E^2 - \mathbf{P}^2 - iqF)\Psi,$$

while reversing the order gives

$$(E - \sigma \cdot \mathbf{P})(E + \sigma \cdot \mathbf{P})\Psi = (E^2 - \mathbf{P}^2 - iqF^*)\Psi.$$

Equation (92) states exactly this: the second-order Klein–Gordon operator is recovered from the square of a first-order object, and the field term carried by F is what remains from the noncommuting covariant momenta.

Free Spin- $\frac{1}{2}$ Factorization

Before the electromagnetic field is turned on, the auxiliary 2×2 map from Eq. (58) already shows how the scalar theory and the two-component field are related. With

$$D_0 := W(i\partial) - m\mathbf{I}, \quad D'_0 := W(i\partial) + m\mathbf{I},$$

one has

$$D'_0 D_0 = -(\square + m^2)\mathbf{I}.$$

Thus every free Dirac solution satisfies the Klein–Gordon equation in each component. Conversely, if Ψ' is a scalar Klein–Gordon solution, then

$$\Psi = \begin{Bmatrix} \Psi' \\ \frac{i}{m} \partial^* \Psi' \end{Bmatrix}$$

with upper component $\Psi' \in \mathbb{C}$ and lower component $\frac{i}{m} \partial^* \Psi' \in \mathbb{C} \oplus \mathbb{C}^3$, solves the free Dirac system. The Dirac equation is therefore a controlled first-order factorization of the same relativistic mass shell rather than a disconnected model.

The Dirac Two-Component System, Gauge Covariance, and Component Form

Define

$$Z = i\partial - q\Phi^* = E - \sigma \cdot \mathbf{P},$$

so that Z is a paravector-valued differential operator with scalar part E and vector part $-\mathbf{P}$. The same map from Eq. (58) then produces a first-order two-component operator whose square is diagonal in the two paravector components. Eq. (94) becomes a first-order system for the two paravector components Ψ_1 and Ψ_2 :

$$(i\partial - q\Phi^*)\Psi_2 - m\Psi_1 = 0, \quad (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 = 0.$$

At the operator level this means

$$\begin{aligned} W(i\partial - q\Phi^*)^2 &= -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi), \right. \\ &\quad \left. (\partial^* + iq\Phi)(\partial + iq\Phi^*)\right). \end{aligned}$$

Multiplying by the mass-conjugate operator $D' = W(Z) + m\mathbf{I}$ gives

$$D' D \Psi = \text{diag}(-E^2 + \mathbf{P}^2 + m^2 - iqF^*, -E^2 + \mathbf{P}^2 + m^2 + iqF) \Psi.$$

Once the square is taken, the coupled Dirac system separates into two Klein–Gordon-type equations whose field terms differ only by the expected conjugation/sign pattern.

The upper and lower components satisfy Klein–Gordon-type equations with opposite field couplings, exactly as recorded in Eq. (95). The sign pattern matters, which is why the caution recorded in Section IX is retained explicitly.

Gauge covariance is then immediate. If

$$\Phi' = \Phi + \partial^* \lambda, \quad \Psi = e^{iq\lambda} \Psi',$$

with $\lambda \in \mathbb{R}$ a real scalar gauge function, then the phase derivative produced by $i\partial$ is cancelled by the shift in the potential, leaving

the form of Eq. (94) unchanged. The supporting tables that follow collect the direct component actions of ∂ , ∂^* , Φ , Φ^* , iqF , and iqF^* ; those tables supply the explicit two-component formulas corresponding to the same field. Tables 32 and 33 record the direct component action of ∂ and ∂^* on a complex paravector test field.

Table 32. Components of $\partial\psi$.

scalar:	$\partial_t\psi_s + \nabla \cdot \psi_v$
vector:	$\partial_t\psi_v + \nabla\psi_s + i(\nabla \times \psi_v)$

Table 33. Components of $\partial^*\psi$.

scalar:	$\partial_t\psi_s - \nabla \cdot \psi_v$
vector:	$\partial_t\psi_v - \nabla\psi_s - i(\nabla \times \psi_v)$

Tables 34 and 35 then show how the potential Φ and its conjugate act on the same field in components.

Table 34. Components of $\Phi\psi$.

scalar:	$V\psi_s + \mathbf{A} \cdot \psi_v$
vector:	$V\psi_v + \psi_s\mathbf{A} + i(\mathbf{A} \times \psi_v)$

Table 35. Components of $\Phi^*\psi$.

scalar:	$V\psi_s - \mathbf{A} \cdot \psi_v$
vector:	$V\psi_v - \psi_s\mathbf{A} - i(\mathbf{A} \times \psi_v)$

Tables 36 and 37 complete the two-component operator tables by recording the field-coupling terms $iqF\psi$ and $iqF^*\psi$.

Table 36. Components of $iqF\psi$.

scalar:	$iq(\mathbf{F} \cdot \psi_v)$
vector:	$iq\psi_s\mathbf{F} - q(\mathbf{F} \times \psi_v)$

Table 37. Components of $iqF^*\psi$.

scalar:	$-iq(\mathbf{F}^* \cdot \psi_v)$
vector:	$-iq\psi_s\mathbf{F}^* + q(\mathbf{F}^* \times \psi_v)$

Low-Energy Scalar and Paravector Limits

To reach the nonrelativistic scalar theory, remove the fast rest-energy oscillation by writing $\Psi' = e^{-imt}\Psi$. Substituting into the minimally coupled Klein–Gordon equation gives

$$0 = (-E^2 - 2mE + \mathbf{P}^2)\Psi,$$

so at leading order in $1/m$,

$$E\Psi \approx \frac{\mathbf{P}^2}{2m}\Psi.$$

Solving for $i\partial_t\Psi$ yields the Schrödinger form

$$i\partial_t\Psi = \left(\frac{(-i\nabla - q\mathbf{A})^2}{2m} + qV \right) \Psi + O(1/m^2).$$

The scalar low-energy limit is therefore already encoded in the relativistic theory.

For the spin- $\frac{1}{2}$ two-component field, one performs the same rest-phase extraction and then eliminates the small component iteratively. At leading order this yields

$$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi + O(1/m),$$

which is Eq. (97). Solving for $i\partial_t\Psi$ gives Eq. (98): the kinetic Schrödinger operator is supplemented by the Zeeman term $-\frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma}$ [9, 17]. The derivation is important because it shows that spin coupling is not appended by hand; it survives as the leading field correction left behind by the relativistic first-order theory.

For a uniform static magnetic field $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$, the projector family from Eq. (45) diagonalizes the Zeeman term:

$$(\mathbf{B} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\|\mathbf{B}\|\Pi(\pm).$$

The decomposition in Eq. (99), namely $\Psi = \Psi(+) + \Psi(-)$ with $\Psi(\pm) = \Pi(\pm)\Psi$, therefore separates the Pauli equation into the two scalar Schrödinger-type equations shown in Eq. (100). The projector identities that make this step possible were derived earlier in Section II, so the Pauli split is a direct physical use of that earlier algebra rather than a new construction introduced only for the wave equation.

VIII. CONCLUSIONS

The algebraic rules fixed in Section II generate interval geometry, covariant transformations, electromagnetism, and relativistic wave equations without changing the underlying language. The later physical equations follow from the earlier algebraic setup rather than appearing as separate formalisms placed side by side.

The main structural point is the complex paravector split introduced in Eqs. (15), (18), and (19). Once a quantity is resolved into real scalar, imaginary scalar, real vector, and imaginary vector parts, the same algebraic product can be read component by component across several parts of physics. In the space-time sections this split yields the interval and proper-velocity formulas. In electromagnetism it packages V with $\mathbf{A}$, and $\mathbf{E}$ with $\mathbf{B}$, into one differential framework. In the wave-equation sections it reappears in the paired field couplings of Eq. (95), where the two component fields carry the conjugate pair $-iqF^*$ and $+iqF$. The benefit is not only compact notation. The same bookkeeping keeps the scalar/vector content visible while also showing which formulas come in natural complementary pairs. Within the flat-space theory developed here, it also avoids explicit tensor notation through the main line of argument.

Electromagnetism is the clearest example. Equation (77) produces the sourced equations from the real scalar and real vector parts, while the complementary imaginary scalar and imaginary vector parts give the magnetic-divergence and Faraday relations in Eqs. (78) and (79). Tables 19 and 20 show that this is not an isolated coincidence: once the full real/imaginary scalar/vector decomposition is kept, the derivative algebra naturally generates partner formulas in which electric and magnetic roles are exchanged. The same pattern appears in the force product. Equation (83) keeps the real part because it gives the observed power and

Lorentz-force terms, while its imaginary companion is

$$\text{img}(FU) = \gamma(\mathbf{v} \cdot \mathbf{B} + (\mathbf{B} + \mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma}),$$

which is the complementary electric-magnetic partner of the real-force expression, with the cross-product orientation adjusted by the algebraic product order.

Not every complementary component carries an accepted physical interpretation. Some are standard and necessary, such as the homogeneous Maxwell equations. Others are better read as algebraically natural partner relations that may encode dual descriptions, hidden symmetries, or possible extensions that have not been fully developed. That distinction matters. The complex scalar/vector paravector framework is rich enough to place established relativistic physics and these additional companion relations inside one common algebraic system. The remaining task is to determine which of these companion formulas are alternate forms of known results and which, if any, carry genuinely new physical content.

The extra companion formulas are therefore worth retaining in view, but they should be interpreted with discipline. Some are already standard parts of accepted theory, while others remain algebraically suggestive rather than physically established.

Remaining Work

The largest missing development is general relativity. The present paper works with a fixed flat-space sigma basis, so curvature never enters. A genuine gravitational extension would require a position-dependent local paravector frame, a covariant derivative with connection terms that describe local rotations and boosts of that frame, and curvature defined from the commutator of those covariant derivatives. In that setting the interval, geodesic motion, Maxwell system, and Dirac system would all have to be rewritten in local frame form, and the energy-momentum discussion around Eq. (85) would have to be replaced by the full source structure needed for Einstein-type field equations. In short, the next gravitational step is not a small correction to the present notation; it is a local-frame version of the whole flat-space construction.

Several other developments are also still missing. The present document does not yet develop non-Abelian gauge structure, so the algebra has not been tested here against the internal symmetries of weak and strong interactions. It does not yet treat many-particle or entangled quantum states, where one needs more than a single paravector or a single 2×2 auxiliary system. It also does not yet give a full account of constitutive laws for anisotropic media, dissipative systems, or continuum mechanics, where higher-rank linear response objects usually appear. On the mathematical side, operator domains, boundary terms, and regularity hypotheses still need to be stated more systematically whenever differential operators are factored or commuted.

Nothing in the present calculations shows a direct incompatibility between STA and the standard relativistic theories treated here. The limitations are instead ones of scope and expressiveness in the current presentation. A single paravector layer does not by itself encode arbitrary higher-rank multilinear structure, bundle

data, or global topological patching. Those features may still be representable in STA, but they require additional constructions beyond what has been developed in this document.

Advantages and Limits of Avoiding Tensors

The main advantage of the present approach is clarity of geometric meaning. The same algebraic product carries scalar contraction, oriented cross-product structure, projector calculus, and transform laws, so many long index derivations collapse to a few lines without losing component information. That compression is not merely cosmetic. It is what makes it possible to place kinematics, electromagnetism, and relativistic wave equations inside one algebraic framework while still recovering the scalar and vector parts explicitly at each step.

The cost is that tensor content has not truly disappeared; it has been repackaged. In flat-space problems that repackaging is often a gain because it removes index bookkeeping and keeps the geometry visible. In curved-space problems the story is more mixed. Once one introduces local frames, connections, curvature, and source maps, the theory again has genuinely multilinear structure even if it is written in STA form. The practical drawback is therefore twofold: first, some problems are not naturally expressed by paravectors alone; second, most of the surrounding literature, software, and standard formulations remain tensor-based, so translation back and forth is unavoidable.

For that reason the tensor-free character of this formulation is best treated as a strong feature, not as a universal replacement principle. In the regime developed here, the algebra gives a compact and coherent reformulation with real conceptual advantages. Beyond that regime, especially in general relativity and in theories built on richer internal or multilinear structure, the more useful question is how much tensor content can be absorbed into the algebra without losing precision, generality, or contact with standard formulations.

REFERENCES

- [1] W. R. Hamilton, “On quaternions; or on a new system of imaginaries in algebra,” *Philosophical Magazine* 25 (1844), 489–495. DOI: <https://doi.org/10.1080/14786444408644923>.
- [2] J. C. Maxwell, “A dynamical theory of the electromagnetic field,” *Philosophical Transactions of the Royal Society of London* 155 (1865), 459–512. DOI: <https://doi.org/10.1098/rstl.1865.0008>.
- [3] W. K. Clifford, “Applications of Grassmann’s extensive algebra,” *American Journal of Mathematics* 1 (1878), 350–358. DOI: <https://doi.org/10.2307/2369379>.
- [4] H. A. Lorentz, “Electromagnetic phenomena in a system moving with any velocity less than that of light,” *Proceedings of the Royal Netherlands Academy of Arts and Sciences* 6 (1904), 809–831.
- [5] A. Einstein, “Zur Elektrodynamik bewegter Körper,” *Annalen der Physik* 322 (1905), 891–921. DOI: <https://doi.org/10.1002/andp.19053221004>.
- [6] H. Minkowski, “Raum und Zeit,” lecture delivered September 21, 1908; published in *Physikalische Zeitschrift* 10 (1909), 104–111.
- [7] O. Klein, “Quantentheorie und fünfdimensionale Relativitätstheorie,” *Zeitschrift für Physik* 37 (1926), 895–906. DOI: <https://doi.org/10.1007/BF01397481>.
- [8] W. Gordon, “Der Comptoneffekt nach der Schrödingerschen Theorie,” *Zeitschrift für Physik* 40 (1926), 117–133. DOI: <https://doi.org/10.1007/BF01390840>.
- [9] W. Pauli, “Zur Quantenmechanik des magnetischen Elektrons,” *Zeitschrift für Physik* 43 (1927), 601–623. DOI: <https://doi.org/10.1007/BF01397326>.
- [10] P. A. M. Dirac, “The quantum theory of the electron,” *Proceedings of the Royal Society A* 117 (1928), 610–624. DOI: <https://doi.org/10.1098/rspa.1928.0023>.
- [11] H. P. Robertson, “The uncertainty principle,” *Physical Review* 34 (1929), 163–164. DOI: <https://doi.org/10.1103/PhysRev.34.163>.
- [12] D. Hestenes, *Space-Time Algebra*, Gordon and Breach, 1966. See also the 2nd edition, Birkhäuser, 2015, DOI: <https://doi.org/10.1007/978-3-319-18413-5>.
- [13] W. E. Baylis, *Electrodynamics: A Modern Geometric Approach*, Birkhäuser, 1999.
- [14] C. Doran and A. Lasenby, *Geometric Algebra for Physicists*, Cambridge University Press, 2003.
- [15] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, 1998.
- [16] W. Greiner, *Relativistic Quantum Mechanics: Wave Equations*, 3rd ed., Springer, 2000.
- [17] J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*, 2nd ed., Cambridge University Press, 2017.

IX. SCOPE NOTES AND CAUTIONS

The list below marks formulas or claims that still require interpretation, correction, or stronger hypotheses than the surrounding exposition alone makes explicit.

1. **The spin-axis commutator estimate in Section II should not be confused with the full Robertson or Robertson–Schrödinger state-variance inequality.**
2. **Any diagonalized matrix form of ds^2 must retain explicit Lorentz-signature information; the Euclidean shorthand $ds^2 = dx' \cdot dx'$ is not a Lorentzian identity.**
3. **In quaternion intermediate expansions, the linear term carries the factor a , as in Eq. (54).**
4. **The line $T = \frac{1}{2}FF^R$ should be read as a frame-conditioned energy-density/Poynting paravector, not the full electromagnetic stress-energy tensor.**
5. **Equation (102) is an on-shell projector statement for normalized free states, not the general Dirac equation.**
6. **An earlier sign variant of Eq. (95) with $+iqF^*$ in the upper component is inconsistent with the operator ordering in $W(i\partial - q\Phi^*)$; the corrected sign is $-iqF^*$.**
7. **Several operator equalities stated in differential form implicitly assume smooth test functions and benign boundary behavior; those hypotheses should be stated whenever commutators of differential operators are used rigorously.**

X. VERIFICATION AND FORMULA STATUS

This section compares the principal statements with standard literature and with the symbolic checks listed immediately afterward, and it distinguishes established reformulations from the smaller set of statements that still require explicit qualification.

STA statement	Literature correspondence	Status	Notes
Eq. (31)	Pauli/GA product split [12, 14]	✓	Standard scalar+bivector decomposition.
Eq. (60)	Minkowski interval in paravector form [12, 13]	✓	Equivalent to $dt^2 - d\mathbf{r}^2$.
Eq. (77)	Compact Maxwell equation in STA [2, 12, 13, 15]	✓	Sign depends on the chosen $\mathbf{F}$ and metric conventions.
Eq. (83)	Lorentz force law in STA packaging [12, 15]	✓	Combines power and force components into one paravector equation.
Eq. (86)	Klein–Gordon equation [7, 8]	✓	Standard free relativistic scalar wave equation.
Eq. (89)	Minimal-coupling KG equation [15, 16]	✓	Operator-domain assumptions should be stated when commuting derivatives and fields.
Eq. (94)	Dirac equation in a 2×2 STA auxiliary representation [10, 12, 14]	✓	Representation-dependent, but equivalent in content once the map to the ordinary Dirac component form is fixed.
Eq. (95)	Squared Dirac system after STA factorization [10, 12, 14]	✓	The upper component carries $-iqF^*$ and the lower component carries $+iqF$ with the operator ordering used here.
Eq. (98)	Leading-order Pauli reduction [9, 17]	✓ with caveat	The displayed form is low-energy and omits higher-order electric-field terms.
Eq. (85)	Electromagnetic energy-momentum density	Needs qualification	Useful frame-dependent object, but not the full stress-energy tensor alone.
Spin-axis commutator estimate	Robertson–Schrödinger uncertainty relation [11]	Qualified	Section II records only the operator-level noncommutativity, not the full state-variance inequality.
Matrix diagonalization remark for ds^2	Diagonalization of indefinite quadratic form	Qualified	Diagonalization reorganizes the quadratic form but does not remove the Lorentzian signature signs.
Eq. (102)	Free on-shell projector condition	Qualified	It is an on-shell projector relation for normalized free states, not the general Dirac dynamics.
Eq. (54)	Quaternion/STA bridge [1, 3]	✓	The corrected linear term carries the factor a , as displayed.

The SymPy verification report below lists the formulas checked directly in code. Its green check marks indicate verified identities, and the comment lines in the $\text{\LaTeX}$ source point to `tools/verify_sta_results.py`.

Python Verification of Main Algebraic and Operator Identities

The identities below were checked symbolically in Python using SymPy and the standard Pauli-matrix realization of the sigma basis. Algebraic identities were reduced directly in matrix form, while differential identities were evaluated on generic test functions $\psi(t, x, y, z)$ with symbolic potentials and field components. The verification script is `tools/verify_sta_results.py`.

Check	Status	Detail
Eqs. (1)–(2)	✓	All nine basis products satisfy $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$ in the Pauli realization.
Eq. (31)	✓	The product $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$ reduces to $\mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$ for symbolic components A_n, B_n .
Eqs. (32)–(37)	✓	The complex paravector product, the Minkowski-type form $ZZ^{*\mathbf{R}} = S^2 - \mathbf{V}^2$, and the Euclidean scalar part $\text{scalar}(ZZ^{\mathbf{R}}) = S ^2 + \ \mathbf{V}\ ^2$ all reduce correctly.
Eqs. (45), (61), (62)	✓	The idempotents $\Pi(\pm)$ satisfy the projector identities, $UU^{*\mathbf{R}} = 1$, and $E^2 - \mathbf{P}^2 = m^2$ follows from $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$.
Eqs. (75), (76), (77)	✓	The potential identity gives $S = \partial_t V + \nabla \cdot \mathbf{A}$, $\mathbf{E} = -\partial_t \mathbf{A} - \nabla V$, and $\mathbf{B} = \nabla \times \mathbf{A}$, while ∂F splits into Gauss, Faraday, and Ampère–Maxwell component equations with the manuscript’s signs.
Eqs. (82) and (83)	✓	The gauge law $\Phi' = \Phi + \partial^* \lambda$ gives $V' = V + \partial_t \lambda$ and $\mathbf{A}' = \mathbf{A} - \nabla \lambda$, and the product FU splits into the expected power term $\gamma \mathbf{v} \cdot \mathbf{E}$ and force term $\gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
Eqs. (90), (91), and (92)	✓	Acting on a generic test function reproduces the expanded Klein–Gordon operator, the commutators $[E, P_m] = iqE_m$ and $[P_m, P_n] = iq(\partial_m A_n - \partial_n A_m)$, and the factorization $(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$.
Eqs. (95), (97), and (100)	✓	Eliminating one Dirac component yields the two squared equations; with the manuscript’s operator ordering, the upper component carries $-iqF^*$ after everything is moved to the left-hand side, while the lower component carries $+iqF$. The exact Pauli identity $(\boldsymbol{\sigma} \cdot \mathbf{P})^2 = \mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}$ holds on a test function; and for uniform $\mathbf{B} = B_0 \mathbf{u}$ the projectors satisfy $(\mathbf{B} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm B_0 \Pi(\pm)$.
Eq. (54)	✓	The quaternion correspondence satisfies $QQ^{\mathbf{R}} = a^2 + \mathbf{b}^2$ and $Q^2 = a^2 - \mathbf{b}^2 - 2ia \mathbf{b} \cdot \boldsymbol{\sigma}$; the omitted factor of a in the older intermediate line is indeed an error.

These symbolic checks complement the literature comparison in Section X. They separate purely algebraic and operator-theoretic identities from statements whose interpretation still depends on physical conventions.